Engineering Science and Technology, an International Journal 77 (2026) 102353

Contents lists available at ScienceDirect

Engineering Science and Technology, an International Journal

journal homepage: www.elsevier.com/locate/jestch

Full Length Article

Enhancing search and rescue missions for victim detection using a coordinated bio-inspired dual-altitude UAV swarm framework

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ARTICLE INFO

Keywords:
Search and rescue
Unmanned aerial vehicles
Small-object detection
YOLO
Unmanned aerial vehicle
Gazebo simulation

ABSTRACT

Robotic systems have become essential tools in disaster response, enabling rapid, safe, and persistent operations in environments that are often inaccessible or hazardous to humans. Among these systems, unmanned aerial vehicles (UAVs) play a particularly critical role in search and rescue (SAR) missions, providing wide-area surveillance, rapid deployment, and real-time situational awareness. Several recent studies have addressed the challenges of multi-UAV coordination and multi-modal search strategies in SAR operations. However, these approaches primarily focus on optimizing UAV trajectories across discretized altitude levels, without explicitly modeling biologically inspired behaviors or integrating real-time altitude-role specialization into sensing and decision-making. To overcome the above issues, this work proposes the development and testing of a coordinated multi-drone approach to achieve efficiency and robustness in such missions. The framework utilizes particle swarm-based path planning to synchronize high-altitude surveying with low-altitude refinement, coordinating thermal and visual sensors to enhance classification performance. Four scenarios are presented, varying in complexity levels, ranging from the simplest, such as detecting humans, to scenes with animal-like thermal signatures for the targets. The method has proven effective in achieving fast coverage, effective trajectory distribution, and stable victim detection. In particular, the low-altitude refinement enhanced the localization precision, while the hybrid sensing was capable of alleviating false positives. The mission time was increased in cases where the environment is more complex, but the overall accuracy and robustness are significantly enhanced. The conclusion is that the coordinated multi-drone approach has great potential for developing SAR missions. Furthermore, the work demonstrates that bio-inspired multi-UAV coordination, combined with multimodal sensor fusion, can significantly enhance the scalability, reliability, and spatial accuracy of SAR operations. The presented framework provides an excellent foundation for future implementation in the context of convention-based disaster response missions.

1. Introduction

The increasing prevalence of natural disasters (for instance, earthquakes, floods, and wildfires) and technological or human-made disasters, such as industrial disasters, chemical disasters, and structural failures, has highlighted the need for effective and efficient search and rescue (SAR) operations. In complex circumstances, the dilemma of locating survivors amidst vast and often hazardous environments poses a challenge, and detection delays are directly correlated with increased casualty rates. In this context, conventional SAR methods, including the use of manned helicopters, ships, and ground teams, are hindered by prolonged response times, limited accessibility in the affected areas, significant operational expenses, and risks to rescuers [1].

Unmanned aerial vehicles (UAVs) are increasingly viewed as an effective and safe alternative to SAR missions, offering rapid deployment, aerial coverage, and the capability for real-time sensing [2].

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https://doi.org/10.1016/j.jestch.2026.102353
Received 23 November 2025; Received in revised form 5 February 2026; Accepted 16 March 2026
Available online 31 March 2026
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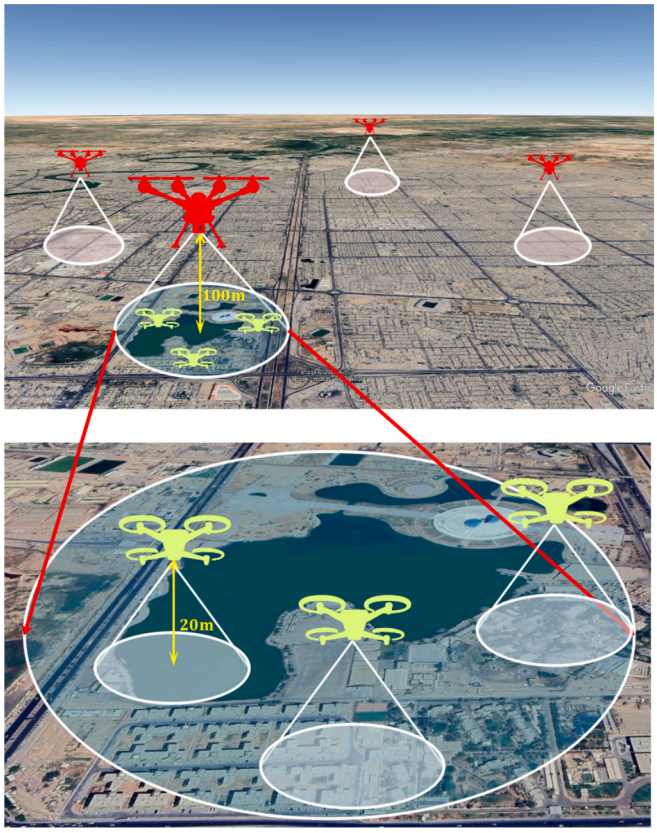

Fig. 1. High and low altitude searching cooperations.

Particularly, UAVs equipped with thermal cameras, LiDAR, or optical sensors can immediately survey hazardous terrain, such as fallen buildings or dense jungles, without putting human life at risk [2,3]. In addition to sensor integration, bio-inspired metaheuristic algorithms have been widely applied to UAV search and motion control due to their adaptability and efficiency in complex, multi-objective conditions. Particle Swarm Optimization (PSO), for example, has been utilized to optimize paths for UAV coverage, demonstrating better convergence and reduced path redundancy compared to deterministic planning based on deterministic grids [4]. Moreover, Grey Wolf Optimization (GWO) and the associated swarm-intelligence algorithms have also been studied for cooperative multi-UAV SAR missions, demonstrating their robustness in target search under uncertainty [5]. In this regard, more recent work suggests the adoption of hybrid methods that combine bio-inspired exploration with task allocation conducted at multiple levels, achieving better scalability in vast search regions [6]. The outcomes make bio-inspired optimization a bright prospect for adaptive UAV coordination under graded search landscapes. However, detection reliability is still an issue. The authors in [7] proposed a digital twin framework tailored for UAV-based wireless channel modeling in 6G "green IoT" systems. Their approach utilizes high-fidelity virtual replicas of aerial-ground communication links to predict channel conditions and optimize UAV deployment strategies. Additionally, the researchers in [8] proposed a cooperative multi-UAV framework that combines adaptive trajectory planning and distributed sensing for exploring dynamic environments. Their method emphasizes inter-agent information sharing and energy-aware path optimization under communication constraints.

Thermal detection is highly effective in disclosing the presence of a human warm signature, even at great heights or under poor visibility, as evidenced by Fig. 1, but is prone to false positives from hot objects or background heat [8]. Conversely, visible-spectrum object detectors, such as YOLO (You Only Look Once), excel at recognizing humans but struggle when deployed from high altitudes due to the small target size and reduced resolution [9]. In this respect, recent studies propose model optimization to enhance YOLO detection on aerial platforms [10]. Yet, none of these methods directly combine the complementary strengths of thermal and visible detection in a hierarchical search strategy for victim detection in SAR missions. Efficient exploration strategies are also vital. Grid-based coverage path planning (CPP) and its variants, such as the hierarchical travelling salesman problem (HTSP) or multi-region planning, have been developed to reduce redundancy and optimize coverage [11,12]. Multi-resolution or multi-altitude frameworks, where UAVs adjust altitude for a coarse overview, followed by fine-detail scanning, have proven effective in resource optimization, such as logistics planning or emergency response scenarios [13]. Furthermore, recent advances in multimodal deep learning have demonstrated the potential of attention-based fusion mechanisms and knowledge distillation techniques to enhance the reliability of perception under conditions of limited or noisy data. For instance, feature-level distillation using soft-hard label paradigms has proven effective for robust place recognition [14], while broader reviews emphasize the role of multimodal integration across navigation and recognition tasks [15]. Additionally, attention-driven multimodal reinforcement learning has shown promise in enhancing situational awareness for intelligent agents [16]. These insights open avenues for future extensions of our framework, where advanced deep learning–based fusion strategies may be incorporated to improve the joint interpretation of thermal and visual data streams in UAV-based SAR operations. In addition, altitude-dependent sensing characteristics further justify hierarchical approaches: low-altitude flights enhance sensor resolution and mapping accuracy [17].

Recent advancements in UAV-based victim detection have expanded beyond traditional full-body recognition by incorporating specialized modalities such as body-region detection and acoustic signal analysis. For instance, the authors in [18] proposed a high-altitude visual recognition framework that leverages YOLOv5 combined with Haar-cascade classifiers to detect discrete human body parts (e.g., head, torso, and limbs), thereby enhancing detection performance in scenarios where only partial body visibility is achievable due to occlusions or distance. In parallel, audio-based detection approaches have emerged as viable alternatives, particularly in low-visibility or cluttered environments. Moreover, the authors in [19] introduced a GAN-based audio denoising pipeline designed for UAV-mounted microphones, effectively suppressing propeller and ambient noise to isolate human voice signals. A subsequent study by the same authors [20] further validated the feasibility of UAV-driven acoustic sensing for locating trapped victims, highlighting the relevance of auditory modalities in SAR contexts.

Recent studies have explored multi-UAV coordination and multi-modal sensing strategies for SAR missions. For instance, [21] presented an evolutionary planning framework capable of configuring UAV searches across multiple altitude modes (K search levels) based on terrain characteristics. However, such approaches primarily focus on global optimization over pre-planned altitude options, with limited attention to biologically inspired hierarchical coordination or real-time altitude-role coupling. In contrast, this work proposes a dual-altitude UAV framework explicitly inspired by animal foraging behaviors, integrating coarse-to-fine search logic with altitude-assigned roles—thermal sensing at high altitudes and visual confirmation at low altitudes—within a unified swarm optimization architecture. This biologically grounded decoupling and role-based coordination introduces a

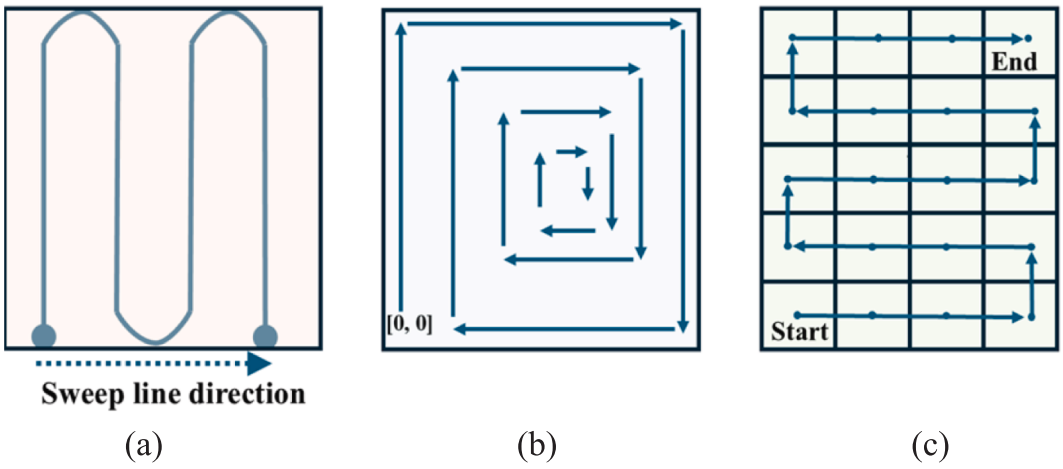

Fig. 2. Traditional searching strategies:(a) Sweep line search; (b) Spiral search; (c) Partitioned search area.

new dimension to multi-UAV SAR systems beyond standard modality or altitude planning. The primary contribution of this study is the formulation and implementation of a bio-inspired dual-altitude search strategy (BSODS) that draws upon natural foraging behaviors to optimize victim localization in disaster environments. The proposed framework integrates swarm-based path planning with a hierarchical altitude structure, enabling coordinated interaction between high-altitude UAVs for broad thermal scanning and low-altitude UAVs for precise visual confirmation. BSODS fuses both tasks of path planning and detection simultaneously, through a biologically inspired optimization layer that guides UAV motion based on thermal cues and swarm dynamics. This integration (altitude-layered coordination, real-time probabilistic victim mapping (PVM), and swarm intelligence) constitutes the principal innovation of the study. The framework's efficacy is demonstrated through comparative simulations against both traditional and bio-inspired planners across diverse SAR scenarios. The main contributions of this paper are as follows:

- A dual-altitude hierarchical UAV search methodology, integrating thermal-based coarse scanning with YOLO-based fine confirmation for robust human localization.
- A grid subdivision and refinement strategy that reduces redundant coverage while improving detection accuracy in candidate regions.
- Integration of bio-inspired swarm algorithms to guide UAV motion in graded grid areas for adaptive and efficient coverage.
- A comprehensive evaluation comparing the proposed approach against single-altitude strategies and thermal-only detection.

The remainder of this article is organized as follows. Section 2 provides an overall review of current UAV-based SAR methods, including the development of the modalities for the sense modality, bio-inspired path planning methods, and object detection methods. Section 3 describes the conceptual bio-inspired dual-altitude UAV search platform, including its modular composition, detection and sense pipeline, as well as PSO-based motion control. Sections 4 and 5 illustrate the simulation configuration settings and experimental scenes implemented to test the system's performance. Section 7 presents the result report, discussing the performance results under different mission complexities where coverage efficiency, detection reliability, and classification accuracy are examined. Finally, Section 8 concludes the paper by highlighting the key contributions, potential real-world deployments, as well as future development and research work.

2. Literature review

2.1. UAVs in SAR missions

UAVs have increasingly featured in SAR missions due to their rapid deployability, broad coverage, and ability to access dangerous sites without endangering human life. UAVs, compared to manned aircraft and ground-based SAR approaches, offer affordable and versatile solutions, enabling real-time situational awareness and scalability in multi-agent systems [22,23]. The latest surveys highlight the importance of integrating optical, thermal, and LiDAR sensors as the prime enablers for UAV-based SAR, permitting enhanced robustness under various disaster circumstances [24]. In this regard, thermal cameras are particularly effective in nighttime or low-visibility environments but are prone to false alarms from background objects that emit heat. On the other hand, visible sensors offer semantic information but perform poorly under low-lighting conditions [25]. In a recent work, the researchers in [26] investigated a multi-UAV SAR mission in complex terrain, combining thermal sensing with real-time path planning across heterogeneous drones. They proposed a probabilistic “hot–zone” map to drive UAV allocation and apply a mobile agent reinforcement learning algorithm to optimize coverage under time constraints. However, they relied on single-altitude operation and lack both hierarchical thermal-to-visual confirmation and *meta*-heuristic-based next-cell coordination. In another relevant study, the authors in [27] presented a multi-UAV cooperative framework for dynamic coverage path planning using ant colony and greedy heuristics in post-disaster zones. They showcased how UAVs share coverage maps and task re-assignments in the presence of communication interference and partial map updates. Their focus was on communication robustness and formation management, but they did not address thermal-based coarse scans or semantic visual confirmation using deep detectors. In an attempt to mitigate the above downfalls, the scientific fraternity has investigated the multi-UAV and multi-modal frameworks where an array of drones collaborate by mutually utilizing the complementary modalities [28]. Furthermore, hierarchical or multi-resolution approaches have been proposed, in which high-altitude UAVs conduct coarse surveys while low-altitude UAVs refine detection

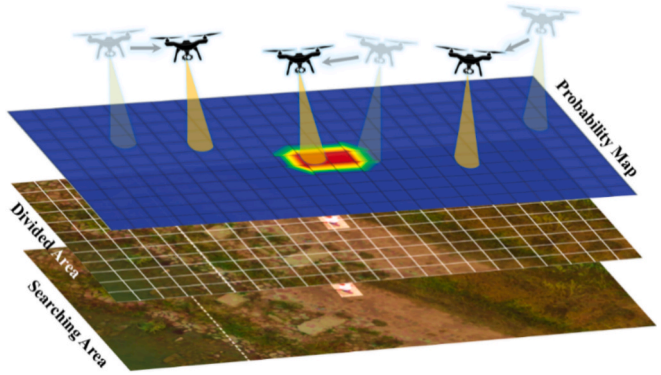

Fig. 3. Intelligent bio-inspired UAV-based SAR system.

with higher resolution and accuracy [29]. In particular, the operational effectiveness of UAVs in SAR missions is increasingly reinforced by advancements in motion control architectures and object detection methodologies. Specifically, robust motion control systems enable UAVs to maintain stability and precise maneuverability under adverse environmental conditions, including strong winds, occlusions, and cluttered terrains, thereby ensuring reliable flight performance and systematic path execution in disaster-affected areas. Simultaneously, object detection technologies have advanced to accommodate the detection of survivors, dangers, and key infrastructure, consequently improving situational awareness and decision-making procedures during missions. The detection processes are carried out either onboard the UAV platform or off-board (on the ground). On-board processing entails the capability for detection with a timely response, an importance that is particularly evident during time-constrained rescue missions. However, it places very high computational requirements on embedded electronics, consequently decreasing flight endurance due to excessive energy consumption. In contrast, off-board processing capitalizes on the powerful ground-station computational capability, allowing for deepened and accelerated analysis while conserving UAVs' energy reserves, consequently increasing mission longevity at the cost of delayed response. Overall, the implementation of sophisticated motion control and detection technologies is a key factor in making UAV-based SAR operations increasingly efficient and reliable.

2.2. UAVs path planning methods for SAR operations

Path planning is an indispensable component of UAV-based SAR missions, as the path planning process directly determines coverage efficiency, mission time, and detection capability. To this end, a series of strategies has been proposed over the last decade, ranging from deterministic approaches based on prespecified patterns to adaptive methods informed by mathematical algorithms inspired by natural processes. Conventional strategies, commonly illustrated in Fig. 2, rely on prespecified search patterns to systematically cover the mission area. The sweep line search method, for example, follows a left-to-right coverage of the search area, thereby achieving extensive coverage with low algorithmic complexity [30]. On the other hand, the spiral search method expands the search area by concentric circles from an inner prescribed initiatory point, so that the covered area is expanded incrementally [31]. Both algorithms have linear time complexity, are low in implementation intensity, and are appealing for missions where time is a critical factor. A block-based search method is the next level of detail from the above techniques, where the overall region is segregated into individual cells or blocks, dividing them into target-containment/target-free ones based on sensor information [32,33]. By combining this technique with formulations like the travelling salesman problem (TSP), block-based methods enable UAVs to determine the shortest path that only attends to worthwhile blocks, thereby reducing the traversal to the red end and improving mission efficiency [34]. All the same, the above deterministic strategies are unable to cope with dynamics from uncertain or changing conditions, as the very rigidity hinders scalability by limiting the ability of multiple UAVs to work together.

To overcome the drawbacks mentioned above, considerable emphasis has been placed on bio-inspired metaheuristic algorithms, which provide versatile and scalable solutions to multi-objective optimization problems, as illustrated in Fig. 3. Specifically, the strategies benefit from the aggregate intelligence found in biological entities to direct UAV maneuvers under complex settings. Ant colony optimization (ACO), for example, has been utilized to optimize multi-UAV search cooperation where digital pheromone trails direct UAVs toward promising locations of interest. For instance, the work presented in [35] Proposed a rule-based multi-agent colony policy that could exhibit improved coordination and performance in challenging maritime environments. On the other hand, the authors in [36] utilized ACO for multi-target detection missions to exhibit a substantial reduction in the trajectory optimization period, as well as the mission execution time [37]. Furthermore, particle swarm optimization (PSO) and its variants have been widely applied to minimize path lengths, balance exploration and exploitation, and accommodate operational constraints such as UAV maneuverability, collision avoidance, and communication limitations [38]. Although bio-inspired metaheuristics offer greater adaptability than traditional strategies, they are still constrained by several challenges.

More precisely, their reliance on heuristic parameterization and partial environmental assumptions often limits the robustness of their

Table 1
Related works comparison summary.

Ref.	Year	Thermal Detection	Object Detection (YOLO/Deep Learning)	Bio-Inspired Path Planning	Grid Division / Hierarchical Search	Multi-Altitude Strategy
[8]	2023	Yes	Yes	No	No	No
[11]	2023	No	No	No	Yes	No
[12]	2025	No	No	Yes	Yes	No
[21]	2019	No	No	Yes	No	Yes
[29]	2018	No	No	No	Yes	Yes
[5]	2024	No	Yes	No	No	No
[9]	2025	No	No	Yes	No	No
[6]	2024	No	Yes	No	Yes	No
[10]	2024	No	Yes	No	Yes	No
[52]	2025	No	Yes	No	Yes	No
[37]	2021	No	No	Yes	No	No
[36]	2024	No	No	Yes	Yes	No
[53]	2022	No	No	No	Yes	No
[54]	2016	No	No	No	Yes	No
Proposed work	–	Yes	Yes	Yes	Yes	Yes

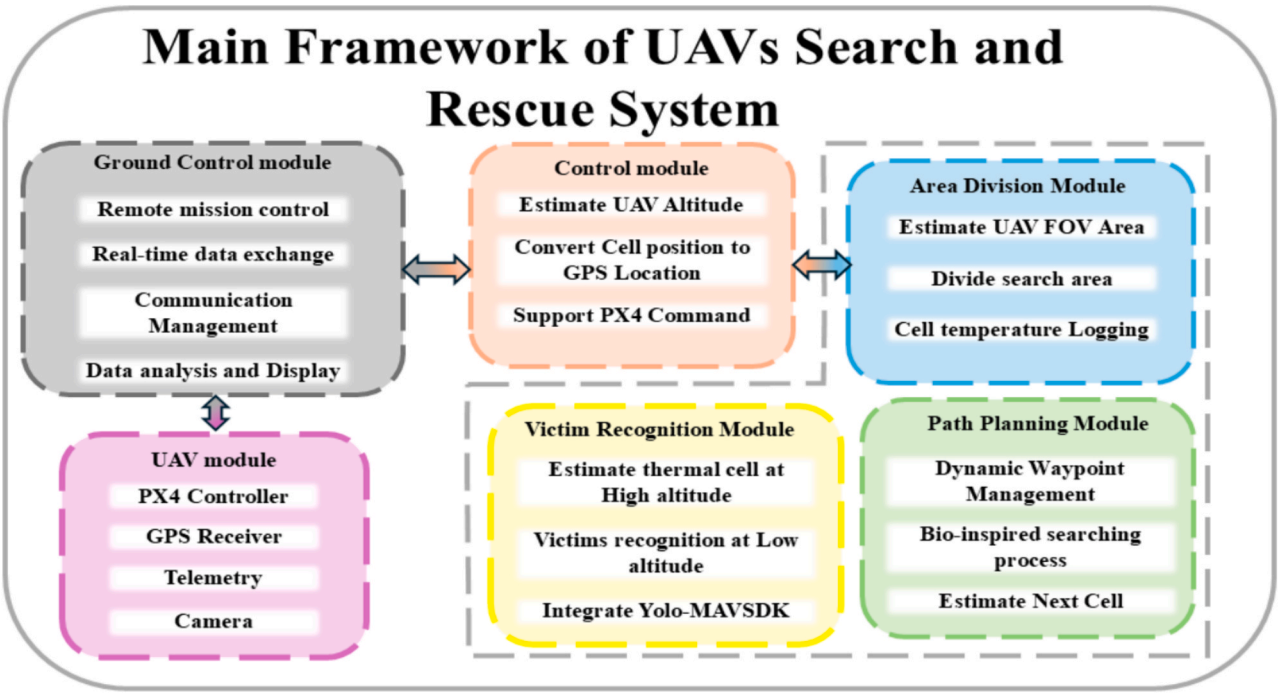

Fig. 4. The proposed modular framework of the UAV-based SAR system.

models in highly dynamic contexts. Moreover, these algorithms are prone to premature convergence and entrapment in local optima, thereby reducing overall search efficiency in large-scale or rapidly evolving environments.

2.3. UAV images object detection

Automated object detection from UAV imagery is fundamental to SAR operations. However, it remains challenging due to the small scale of human targets, background clutter, occlusions, and limited onboard computational resources. In practice, manual inspection of aerial video is inefficient and prone to error, underscoring the need for robust real-time detection algorithms [39,40]. In this context, recent advances in deep learning, particularly the YOLO family of detectors, have significantly improved aerial person detection (APD), combining high inference speed with competitive accuracy [41,42]. However, generic datasets such as COCO [43] are insufficient for UAV SAR, as they lack small-scale, sparsely distributed targets. Dedicated datasets such as VisDrone [44] and HERIDAL [45] address this gap to enable practical model training for UAV-based detection tasks.

To enhance sensitivity to small objects, modern YOLO variants integrate multi-scale feature fusion (FPN, BiFPN, PB-FPN) and attention mechanisms (e.g., CBAM) [46,47]. Specifically, optimized architectures, such as YOLOv5s with PB-FPN and deconvolution layers, have demonstrated superior performance (mAP@50 = 0.802 on Heridal) while remaining suitable for real-time deployment on embedded hardware (Jetson Nano) [9]. Other domain-specific adaptations include YOLOv5 with Kalman filtering for thermal tracking [48], YOLO-IHD for indoor human detection [49], and lightweight designs such as PS-YOLO [50] and YOLO-SRMX [51], all of which emphasize the trade-off between accuracy and computational efficiency. Detection accuracy is also influenced by flight altitude, with performance degrading when operational conditions diverge from the training distribution (e.g., Heridal: 40–65 m) [9].

This dependency is particularly relevant in dual-altitude SAR frameworks, where high-altitude UAVs perform coarse thermal scanning and low-altitude UAVs conduct YOLO-based verification. Accordingly, YOLO-based detectors optimized with domain-specific datasets, feature fusion, and lightweight architectures represent a promising foundation for UAV SAR. Nevertheless, challenges remain regarding dataset bias, generalization to unseen environments, and reliable real-time performance on resource-constrained UAV platforms. Table 1 summarizes key contributions of existing literature in terms of thermal detection, object detection, bio-inspired path planning, grid-based division, and multi-altitude strategies. As shown, prior works have advanced individual components: some emphasize path optimization (e.g., PSO- or GWO-based planning), while others improve detection using YOLO or thermal sensors. However, very few studies attempt to integrate these dimensions. This research gap motivates the present study, which introduces a thermal-driven search, bio-inspired motion control, and fine-grained YOLO verification in a hierarchical UAV framework.

As summarized in Table 1, the distinctiveness of the proposed framework resides in its tight integration of bio-inspired swarm coordination with a dynamically updated PVM derived from thermal imagery. In contrast to prior hierarchical UAV-SAR approaches that rely on pre-planned coverage or static thermal overlays, this study introduces a real-time, adaptive decision-making mechanism in which PVM scores continuously influence the UAVs' movement via PSO. Additionally, the framework establishes a hierarchical sensing architecture, coupling wide-area high-altitude thermal scanning with localized low-altitude YOLO-based semantic confirmation. This dual-modality process enhances mission adaptability, reduces false positives, and supports robust victim localization under complex terrain constraints. Collectively, these innovations represent a methodological advancement beyond existing hybrid UAV SAR systems.

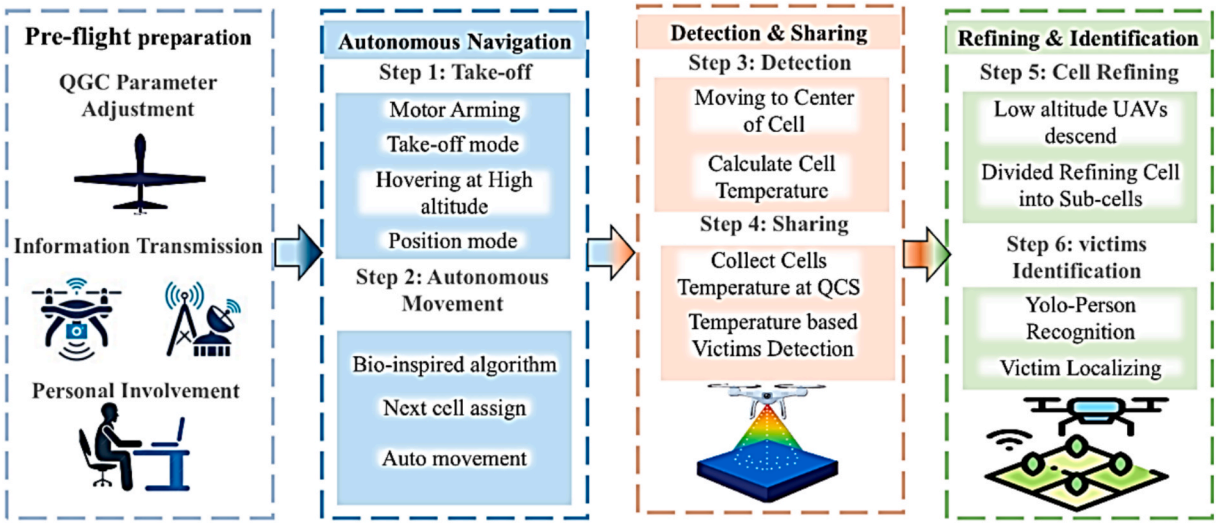

Fig. 5. Operational roadmap of the proposed UAV-based SAR framework.

3. Materials and methods

3.1. The proposed UAV-based searching framework

The proposed UAV-based SAR framework is designed as an integrated and modular system that combines sensing, path planning, recognition, control, and ground-based data management to achieve reliable and efficient victim detection and localization. As illustrated in Fig. 4, the architecture is organized into six interdependent modules that collectively ensure systematic mission execution:

1. UAV Module: Comprising the PX4 autopilot, GPS receiver, telemetry unit, and onboard camera, this module provides the physical and sensing platform for autonomous aerial operations.
2. Control Module: This module is responsible for altitude estimation, the transformation of grid cell coordinates into global positioning waypoints, and the execution of PX4-based flight commands.
3. Area Division Module: This module partitions the mission environment into discrete grid cells by estimating the UAV's field of view (FOV), managing coverage allocation, and logging cell-level thermal data.
4. Victim Recognition Module: This module implements a dual-altitude detection strategy. At higher altitudes, UAVs perform coarse scanning using thermal imaging to identify potential hotspots. In contrast, at lower altitudes, a dedicated UAV conducts fine-grained inspections using YOLO-based person detection.
5. Path Planning Module: This module employs a bio-inspired algorithm to allocate waypoints dynamically, determine the next grid cell to be visited, and optimize multi-UAV trajectories, thereby reducing redundancy and improving exploration efficiency.
6. Ground Control Module: This module provides centralized supervision, enabling remote mission control, real-time data exchange, communication management, and visualization of mission progress through data analysis and display.

Collectively, these modules establish a robust architecture that integrates high-altitude thermal exploration with low-altitude visual verification, ensuring both large-scale coverage efficiency and precise identification of victims.

The operational workflow of the proposed framework is depicted in Fig. 5, which presents the mission roadmap structured into four sequential phases:

- **Phase 1 (Pre-flight Preparation):** UAVs undergo parameter configuration through the Ground Control Station, communication channels are initialized, and operator inputs are incorporated for mission setup.
- **Phase 2 (Autonomous Navigation):** UAVs perform take-off and high-altitude hovering before commencing systematic exploration. Their trajectories are dynamically governed by a bio-inspired optimization algorithm, which enables adaptive and scalable coverage of the search area.
- **Phase 3 (Detection and Sharing):** UAVs record thermal measurements at the center of each grid cell, calculate temperature-based indicators of potential victims, and share results with the ground control station. Hotspot cells are subsequently flagged as candidate regions requiring further inspection.
- **Phase 4 (Refinement and Identification):** A low-altitude UAV is deployed to revisit flagged cells. These regions are subdivided into finer sub-cells, within which YOLO-based person detection is employed to confirm human presence and precisely localize victims.

This hierarchical dual-altitude approach leverages the complementary strengths of thermal imaging for wide-area coverage and deep learning-based visual recognition for fine-scale confirmation. By coupling multi-sensor fusion with bio-inspired path planning and adaptive grid refinement, the proposed framework addresses the limitations of existing single-altitude or single-modality SAR systems, offering a reliable and efficient solution for real-world rescue missions.

3.2. UAV autonomous motion control

3.2.1. Dividing the searching area

To ensure systematic and complete coverage of the area by each UAV, the global search space is discretized into grid cells. The grid dimensions are computed based on the UAV's altitude and the intrinsic parameters of the onboard thermal camera, specifically its horizontal and vertical fields of view (FOV). Algorithm 1 formalizes this process

and defines the mathematical relationship between the FOV, the altitude, and the grid cell dimensions.

Algorithm 1: Dividing the Search Area into Grid Cells

Inputs:
- Aw: Total width of the searching area in meters
- Ah: Total height of the searching area in meters
- H: Altitude of the drone in meters
- fov_x: Horizontal field of view of the camera (degrees)
- fov_y: Vertical field of view of the camera (degrees)

Outputs:
- Grid: 2D array representing search cells of size $[cells_x][cells_y]$
- $cell_{width}$: Width of each grid cell
- $cell_{hight}$: Height of each grid cell

Function $Divide_Area_into_Grid$(Aw, Ah, H, fov_x, fov_y):
- $fov_x = fov_x * (\pi/180)$ // Convert horizontal FOV to radians
- $fov_y = fov_y * (\pi/180)$ // Convert vertical FOV to radians
- $cell_{width} = 2 * H * tan(fov_x/2)$ // Compute cell width from fov_x and H
- $cell_{hight} = 2 * H * tan(fov_y/2)$ // Compute cell height from fov_y and H
- $cells_x = \frac{A_w}{cell_{width}}$
- $cells_y = \frac{A_h}{cell_{hight}}$
- Grid = $Create_Empty_Grid(cells_x, cells_y)$
- **Return** Grid, $cell_{width}$, $cell_{hight}$

End

This formulation ensures that each grid cell corresponds to the effective ground footprint of the UAV's camera at a given altitude. Consequently, each cell is designed to be fully visible in a single top-down image frame, facilitating coverage assurance and efficient sensor deployment.

3.2.2. Bio-inspired planner

1. Swift flight optimizer

Within the delineated UAV-based search and rescue (SAR) framework, the Swift Flight Optimizer (SFO) has been modified to control the choice of successive grid cells for exploration. The SFO constitutes a bio-inspired metaheuristic that models the collective and adaptive flight dynamics observed in swift birds [55]. In contrast to conventional swarm planners, which are based on a single kinematic model, SFO can provide a biologically based multimodal flight paradigm allowing global exploration and local exploitation to be adaptively reconciled when given real-time mission feedback. SFO is inspired by the adaptive flight behavior of swift birds and has three different modes of behavior: Glide, Target, and Micro, making it particularly suitable for SAR tasks that require both wide-area exploration and fine-grained victim localization. Within the proposed SAR framework, the so-called Glide and Target modalities are concretely instantiated through UAV motion strategies, thereby enabling the hierarchical exploration of the search space in conjunction with victim-centric convergence.

In Glide mode, the velocity and position of the agent i are updated as:

$$v_i^{t+1} = \omega . v_i^t + \alpha . N(0, I) + \beta . \boldsymbol{L} \tag{1a}$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{1b}$$

where v_i^{t+1} is the current velocity, ω is the inertia weight, $\alpha and \beta$ are scaling factors for Gaussian and Lévy components, $N(0, I)$ is a vector of standard normal values. Eq. (2) is used to define Lévy step components L_j

$$L_j = \frac{u_j}{|v_j|^{1/\beta}} \tag{2}$$

where $u_j \sim N(0, \sigma_u^2)$ is a standard normal variable, $v_j \sim N(0, 1)$ and $\beta \in (0, 2]$ is a stability parameter (commonly set to 1.5).

Eq. (3) is used to compute the scale parameter σ_u:

$$\sigma_u = \left[\frac{\Gamma(1+\beta).\sin(\frac{\pi\beta}{2})}{\Gamma\left(\frac{1+\beta}{2}\right).\beta.2^{(\beta-1)/2}} \right]^{1/\beta} \tag{3}$$

where $\Gamma(.)$ is the gamma function. Glide mode induces large exploratory displacements by introducing random perturbations along with Lekley-type variations, making it particularly effective for SAR coverage sce-

narios that involve avoiding redundant scanning while maintaining search diversity. In this study, Glide mode is primarily activated when the inter-UAV spacing decreases (i.e., when drones start to cluster) or when a large number of unvisited or low-confidence cells remain in the mission, allowing the swarm to expand its spatial footprint. When a promising region is detected during the search, Target mode appoints agent motions to be goal-directed attraction towards the best solutions:

$$v_i^{t+1} = \omega . v_i^t + \psi_1 r_1 \left(p_{best} - x_i^t\right) + \psi_2 r_2 \left(g_{best} - x_i^t\right) \tag{4a}$$

$$x_i^{t+1} = x_i^t + v_i^{t+1} \tag{4b}$$

where p_{best} is the personal best position of the agent i, g_{best} is the global best position among all agents, ψ_1 and ψ_2 are acceleration coefficients and r_1, r_2 are random scalars $\in$ [1,2]. Target mode is employed in this thesis to drive rapid convergence toward a victim-suspected region, i.e., a grid cell exhibiting high victim probability (as determined by thermal scoring). Once a candidate cell is identified, Target mode prioritizes exploitation by guiding the assigned UAV(s) toward the most promising coordinates for confirmation.

As a solution at each step t, the search process is represented by (5):

$$x^t = \{x_1^t, x_2^t, \cdots.., x_N^t\} \tag{5}$$

where x_j^t denotes the grid-cell position of the UAV agent $j \ni \{1, 2, 3, \cdots, N\}c$. At each time step, its fitness value is calculated as the ratio of the confidence that a victim is detected in the candidate cell to the time needed to reach and inspect the candidate cell, thereby balancing the probability of detecting a victim with the efficiency of the search. The objective function is defined to maximize the likelihood of detecting victims and minimize the cost of traveling, and is described as (6):

$$f_j\left(x_j^t\right) = \frac{P(x_j^t)}{d(x_j^{t-1} - x_j^t)/v} \tag{6}$$

where $P\left(x_j^t\right) \in$ [1] denotes the thermal score associated with the current cell x_j^t, which represents the probability of the victim's presence, $d(x_j^{t-1} - x_j^t)$ is the UAV traveling the Euclidean distance j from its previous cell x_j^{t-1} to the current cell x_j^t, and v is the constant cruising velocity of the UAV.

The detailed procedure of the SFO-based UAV motion control is summarized in Algorithm 2.

Algorithm 2: SFO-based Multi-UAVs Search Planner

Input: Grid Cells $C_{i,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Assign the personal best P_{best} and the global best G_{best}
While the termination criterion is False, **do**
1 **Foreach** UAV in N do:
2 **Update** mode (Glide / Target) based on rules
3 **Update** velocity and position using the mode formula
4 **Update** the cell position using mode formula
5 **Deploy** the drone to the assigned cell.
6 **Evaluate** the objective function using (6)
7 **If** $f_j\left(x_j^t\right) < f_j(P_{best,j})$:
8 **Update** $P_{best,j}$
9 **End**
10 **If** $f_j\left(x_j^t\right) < f(G_{best,j})$:
11 **Update** $G_{best,j}$
12 **End**
13 **End**
14 **End**

2. Particle swarm optimization

The motion of the UAVs between cells is guided by a bio-inspired path planning algorithm, adapted from the Particle Swarm Optimization (PSO) paradigm. The PSO, initially introduced by Kennedy and Eberhart [56], is one of the most widely used algorithms in swarm intelligence. PSO has been widely adopted in UAV path planning due to its scalability, low computational overhead, and robustness in real-time applications. In the proposed UAV-based SAR framework, PSO is adapted to guide the UAVs in selecting the next grid cell to visit during the search process. Specifically, each UAV is modeled as a particle that explores the discretized grid space according to swarm dynamics, striking a balance between local exploration and global coordination.

To find the position of UAV (j), the next step is determined by (7):

$$x_j^{next} = x_j + v_j^{next} \tag{7}$$

where the velocity vector v_j^{next} is computed as given in (8):

$$v_j^{next} = \omega_{inertia} \cdot v_j + \omega_{cognitive} \cdot r_1 \odot (P_{best,j} - x_j) + \omega_{social} \cdot r_2 \odot (G_{best,j} - x_j) \quad (8)$$

where r_1 and r_2 are uniformly distributed random numbers in [1]. The weights $\omega_{inertia}, \omega_{cognitive}, and \omega_{social}$ control the influence of the previous velocity, the UAV's personal-best cell, and the swarm's global-best cell, respectively.

The detailed procedure of the PSO-based UAV motion control is summarized in Algorithm 3.

For navigation, each selected grid cell centroid (C_i, C_j) is transformed into a GPS waypoint (ϕ, λ) using a geodetic conversion referenced to the mission origin. This step enables direct transmission of the updated waypoints to the UAV autopilot for autonomous motion execution.

3. Ant-colony optimization

Ant Colony Optimization (ACO) is a probabilistic metaheuristic inspired by the foraging behavior of ants [57]. In this strategy, artificial

Algorithm 3: PSO-based Multi-UAVs Search Planner

Input: Grid Cells $C_{x,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Assign the personal best P_{best} and the global best G_{best}
While the termination criterion is False, **do**
1 **For** each UAV in N do:
2 Compute the velocity using (8)
3 Update the cell position using (7)
4 Deploy the drone to the assigned cell.
5 Evaluate the objective function using (6)
6 **If** $f_j(x_j^t) < f_j(P_{best,j})$:
7 **Update** $P_{best,j}$
8 **End**
9 **If** $f_j(x_j^t) < f(G_{best,j})$:
10 **Update** $G_{best,j}$
11 **End**
12 **End**
13 **End**

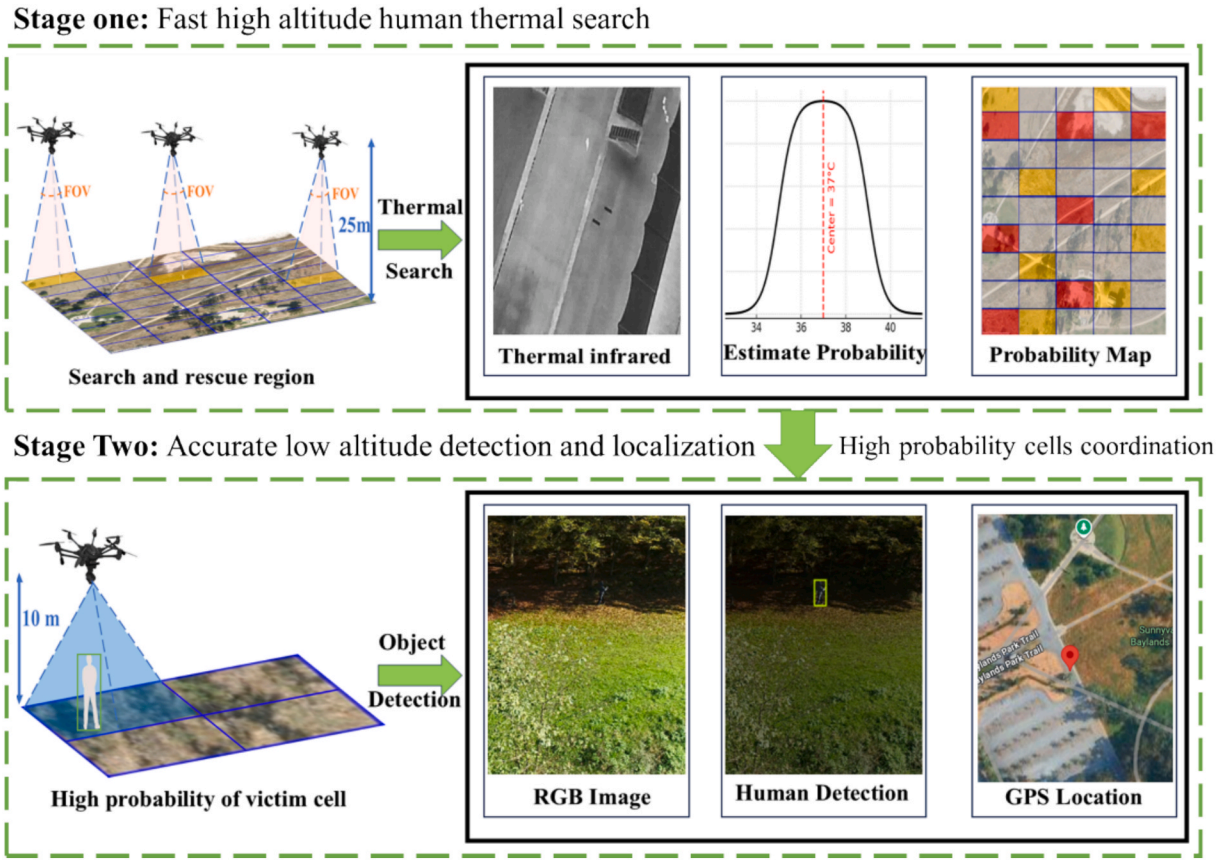

Fig. 6. Two-stage UAV-based victims' detection and localization scheme.

agents (drones) construct paths based on pheromone trails and heuristic information that reflects the quality of the visited regions. In the UAV search context, each drone behaves like an independent ant, selecting its next movement based on a balance between pheromone strength (exploiting known good paths) and heuristic desirability (exploring promising areas). In SAR missions, ACO is particularly useful for ensuring distributed search trajectories and minimizing overlap between UAVs. The probability of selecting the next cell (i,j) from the current position is determined by (9):

$$p_{ij}^{k}(t) = \begin{cases} \frac{\left[\tau_{ij}(t)\right]^{\alpha}.\left[\eta_{ij}(t)\right]^{\beta}}{\sum_{i \ni allowed_i^k}\left[\tau_{ij}(t)\right]^{\alpha}.\left[\eta_{ij}(t)\right]^{\beta}}, if j \ni allowed_i^k \\ 0, otherwise \end{cases} \quad (9)$$

where $\tau_{ij}(t)$ denotes the pheromone intensity associated with the transition from cell i to cell j at iteration t, $\eta_{ij}(t)$ represents the heuristic desirability of selecting cell j, defined as a function of the victim probability or the normalized thermal response, and α and β are the control parameters that balance the influence of pheromone trails and heuristic information, respectively.

After all UAVs complete a round of moves, pheromone trails are updated globally as given in (10):

$$\tau_{ij}(t+1) = (1-\rho).\tau_{ij}(t) + \sum_{k=1}^{m} \Delta\tau_{ij}^{k} \quad (10)$$

where $\rho \ni [0, 1]$ is the pheromone volatilization rate factor, $(1-\rho)$ is the ant pheromone residual factor, and $\Delta\tau_{ij}^{k}$ is the pheromone volatilization rate coefficient of ants k released pheromone.

The detailed procedure of the ACO-based UAV motion control is summarized in Algorithm 4.

Table 2
The specific training hyperparameter information of the model.

Hyperparameters	Value
Learning Rate	0.01
Momentum	0.937
Epochs	300
Batch Size	8
Images Size	640
Optimizer	SGD
Weight Decay	0.0005
Work	8
Pretrained	False
Warmup Epochs	3.0
Warmup Momentum	0.8
Warmup Bias Learning Rate	0.1

Table 3
LEAF-YOLO-N specifications and performance metrics.

Metric	Value
Parameters (M)	1.2
GFLOPs (G)	5.6
Inference speed (FPS)	15 FPS (Jetson nano 4GB)
mAP_{50}	41%
$mAP_{50:95}$	21%

Algorithm 4: ACO-based Multi-UAVs Search Planner

Input: Grid Cells $C_{x,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Initialize $\tau(i,j) = \tau_0$ for all grid cells (i,j).
While the termination criterion is False, **do**
1 **For each** UAV in N do:
2 Compute $P(i,j)$ for neighbor cells using Equation (9)
3 Select the next cell j using the roulette-wheel selection.
4 Deploy the drone to the assigned cell.
5 Evaluate the objective function using (6)
6 **End**
7 Update $\tau(i,j)$ using Equation (10)
8 **End**

4. Cuckoo search

Cuckoo Search (CS) is a metaheuristic optimization algorithm that simulates the brood parasitism strategy of a particular cuckoo species. This method was first proposed by Yang and Deb in 2009 and is widely used in search and rescue, signal processing, and engineering optimization due to its simplicity and global search capability [58]. Its primary strength lies in promoting long-distance exploratory steps, which helps avoid premature convergence. In this framework, CS contributes to enhancing coverage diversity, especially in large-scale or sparsely populated environments where exploration of unvisited regions is critical.

In this technique, each cuckoo of the CS algorithm lays only one egg at a time in a randomly selected nest from F nests. The top (highest-quality) nests (agents' positions) will pass their eggs to the next generation. The fitness value characterizes each individual, and the high-quality egg corresponds to the agent's high fitness value. Moreover, the host-bird finds the cuckoo egg in its own nest with a probability $\rho\alpha\in$[1] and either removes it if the cuckoo eggs that were removed are not of high quality or migrates to another place to make a new nest. The

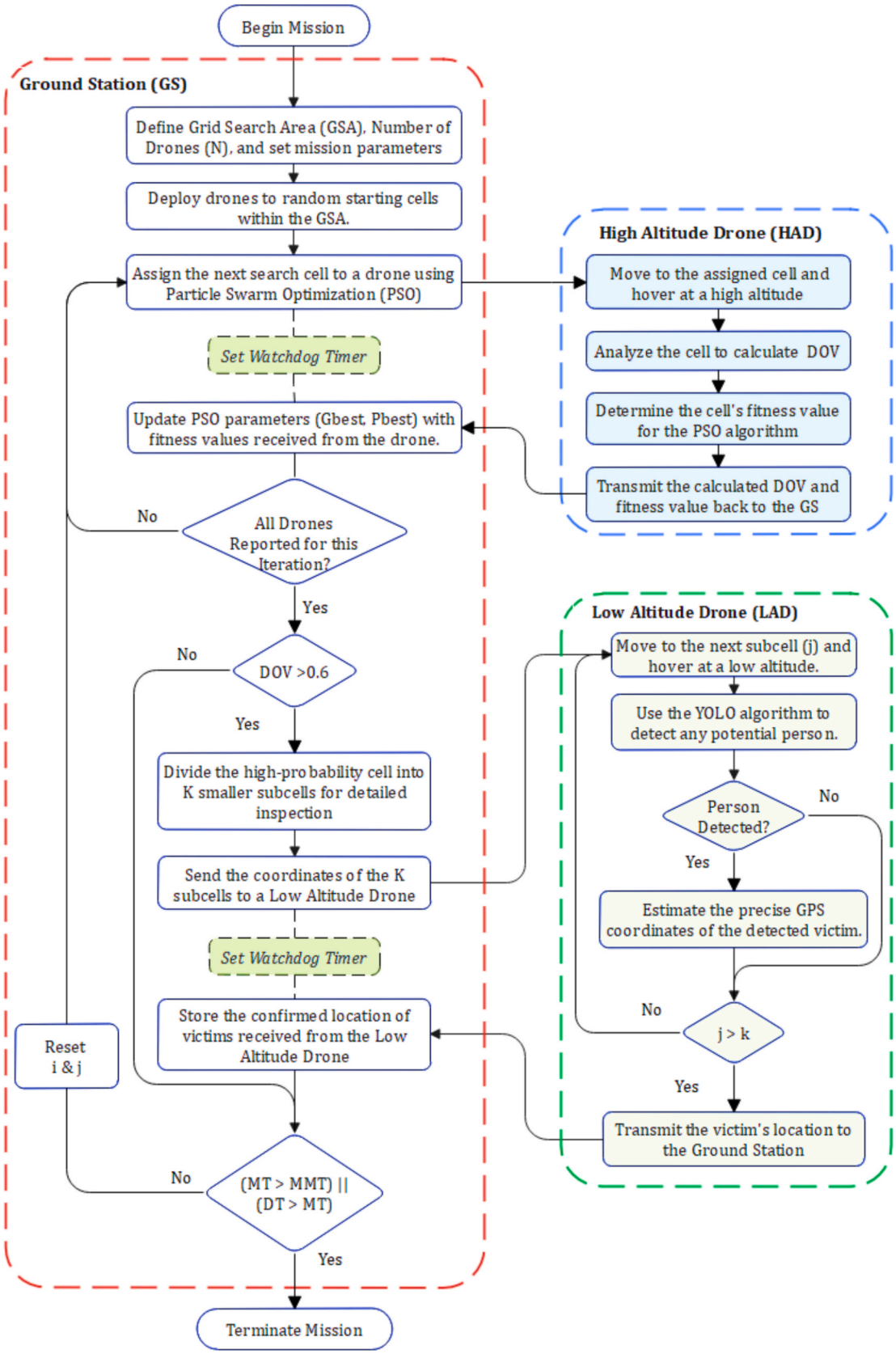

Fig. 7. Flowchart of the proposed bio-inspired dual-altitude UAV search framework.

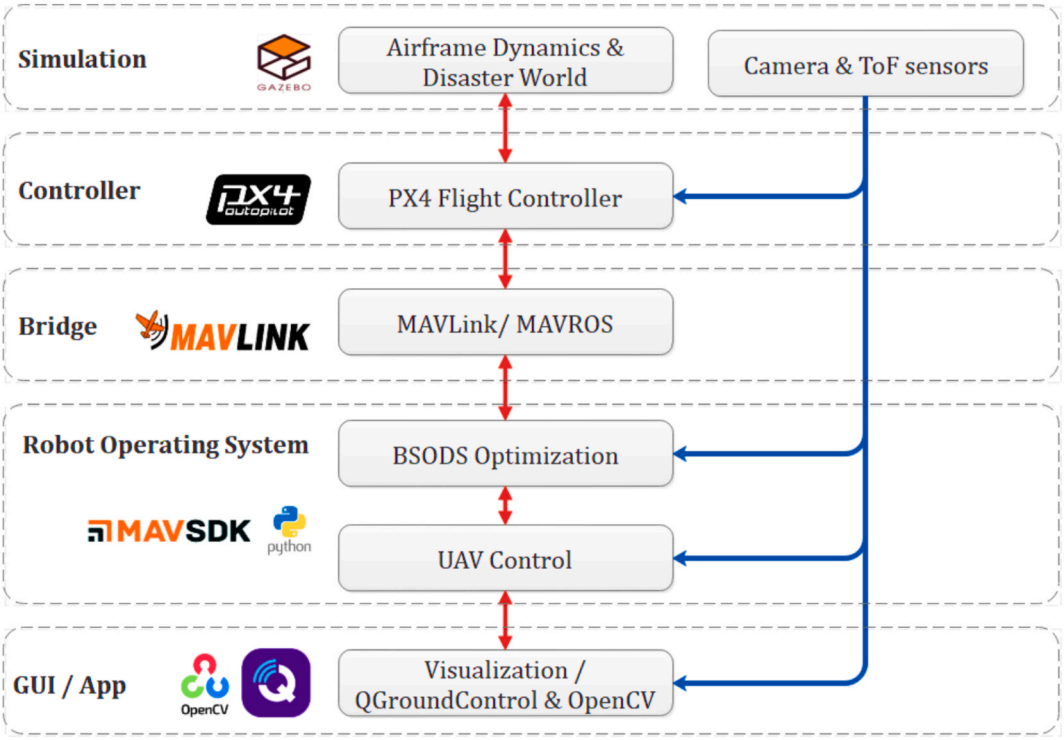

Fig. 8. Workflow and software architecture of the simulation framework for the BSODS strategy.

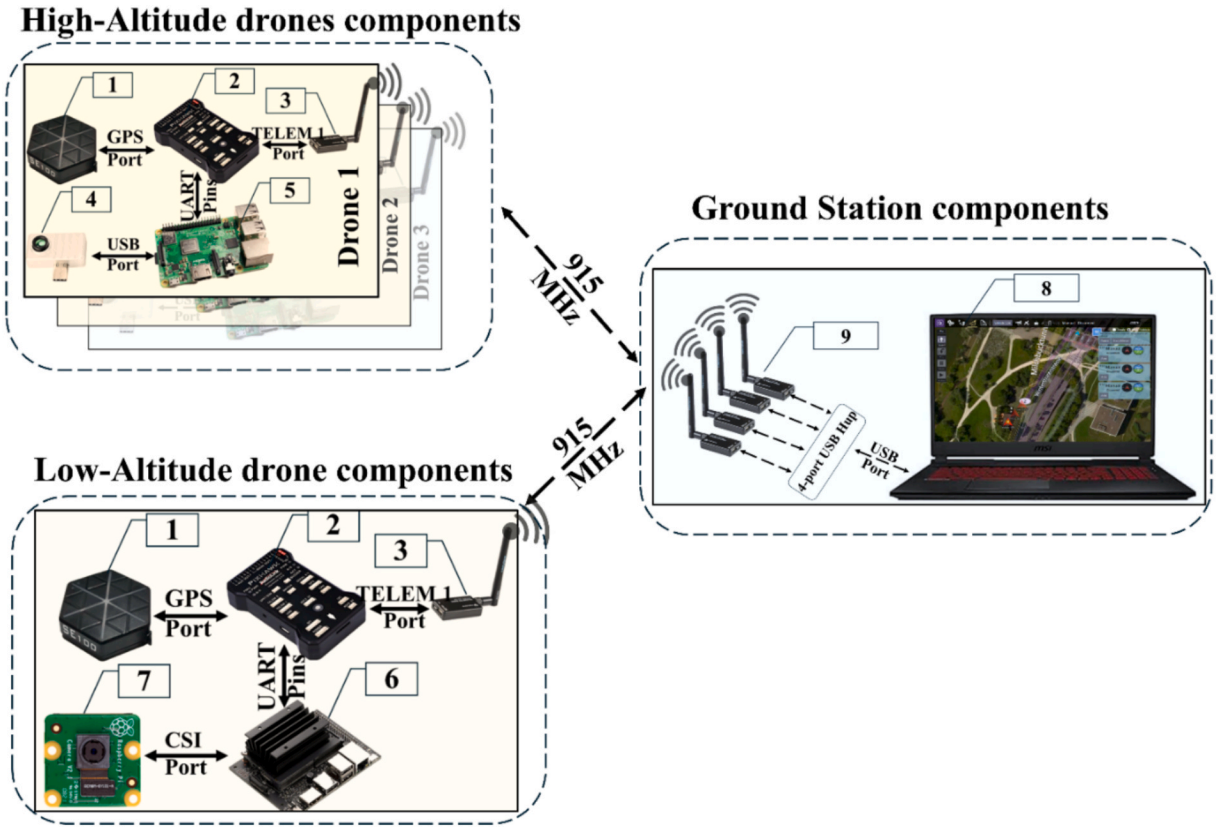

Fig. 9. Hardware structure of UAV-based SAR system: (1) GPS module, (2) Pixhawk FC, (3) Telemetry (Air unit), (4) thermal camera, (5) Raspberry Pi 3, (6) NVIDIA Jetson Nano, (7) telemetry (ground unit), (8) PC ground station.

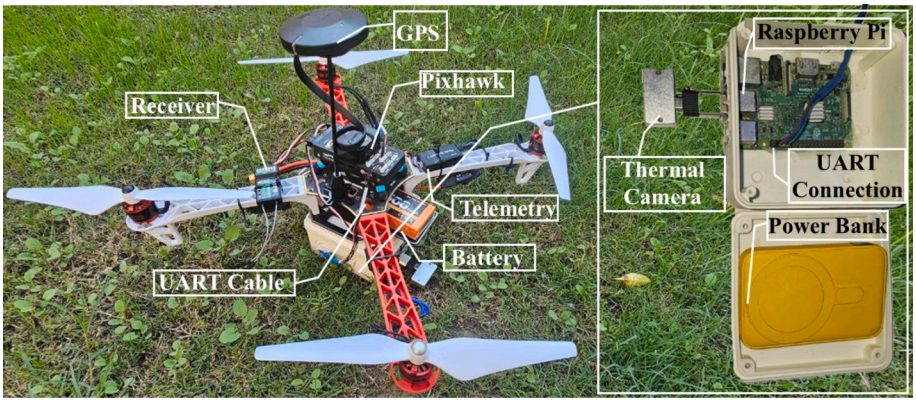

Fig. 10. Picture of a High-altitude UAV searching system.

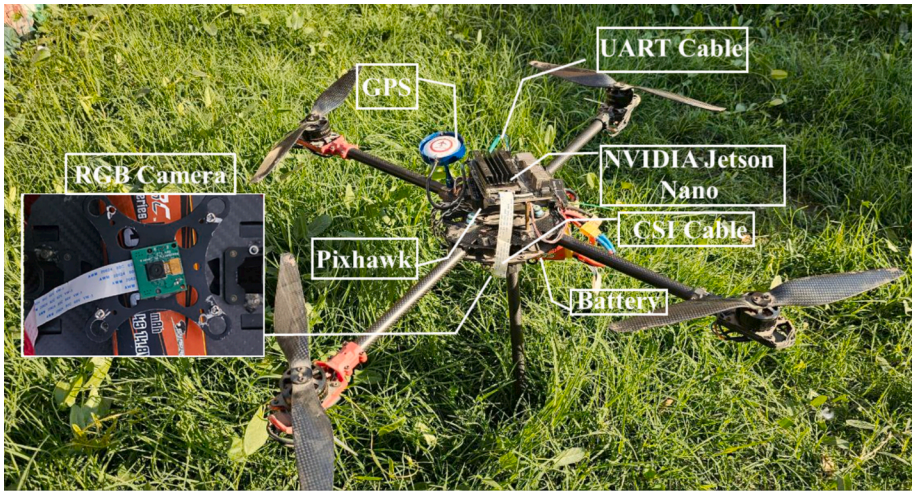

Fig. 11. Picture of a Low-altitude UAV searching system.

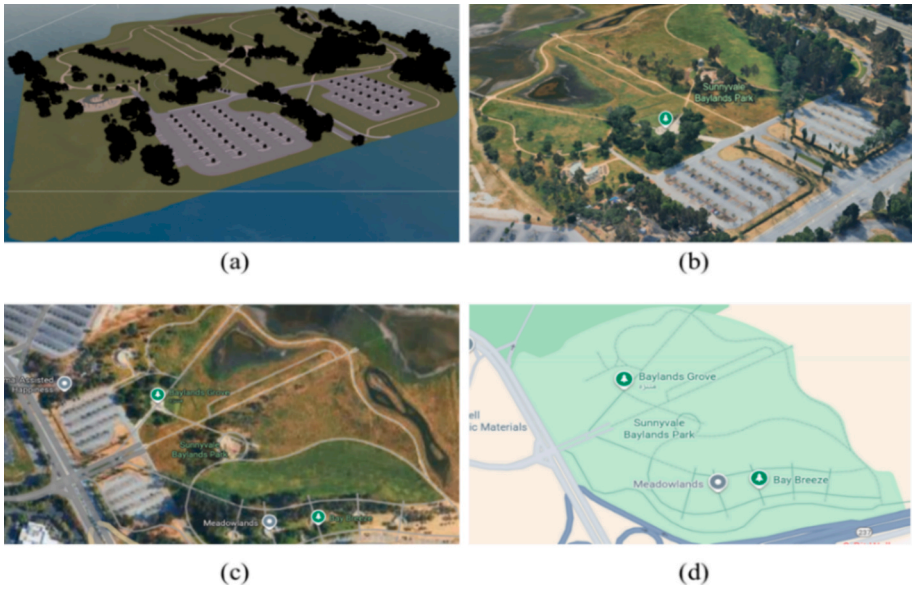

Fig. 12. Gazebo-based simulation environment and its correspondence to the real-world Sunnyvale Baylands Park: (a) Gazebo simulation model, (b) Google Earth 3D perspective, (c) Google Maps satellite imagery, and (d) map representation.

CS algorithm position update formula is also nonlinear.

The movement of each agent j using the CS algorithm is given by the following formulas (11):

$$x_j^{t+1} = x_j^t + a_s \odot \text{Levy}(\lambda) \tag{11}$$

where $as > 0$ is the step size, proportional to the problem's scale, and Levy (λ) is the Lévy step defined as given in (12):

$$\text{Levy}(\lambda) = \frac{u_j}{|v_j|^{1/\beta}} \tag{12}$$

where $u_j \sim N(0, \sigma_u^2)$ is a standard normal variable, $v_j \sim N(0,1)$ and $\beta \in (0,2]$, which is a stability parameter (commonly set to 1.5). The scale parameter σ_u is calculated using (13):

$$\sigma_u = \left[\frac{\Gamma(1+\beta).\sin(\frac{\pi\beta}{2})}{\Gamma\left(\frac{1+\beta}{2}\right).\beta.2^{(\beta-1)/2}} \right]^{1/\beta} \tag{13}$$

where $\Gamma(.)$ is the gamma function. The CS procedure is described in Algorithm 5.

Algorithm 5: CS-based Multi-UAVs search planner

Input: Grid Cells $C_{x,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Initialize a population of n nests with random grid-based paths.
While the termination criterion is False, **do**
1 **For** each UAV in N do:
2 Generate new nest position j via Lévy flight according to (11);
4 Deploy the drone to the assigned cell.
5 Evaluate the objective function using (6)
6 **If** $f_j(x_j^t) < f(x_j^{t-1})$:
7 Replace x_j^{t-1} with the newly generated solution.
8 **End**
9 Keep the best solution.
10 **End**
11 **End**

3.2.3. Traditional planners

1. Breadth-first search

Breadth-First Search is a well-known and commonly used graph traversal algorithm that visits all neighboring nodes at the current level of depth before moving on to the nodes at the next level [59]. This algorithm ensures finding the shortest path (the shortest number of steps) in an unweighted graph, and it is beneficial when the environment is discrete, divided into grid cells. At each decision point, the UAV selects the nearest unexplored neighbor cell to its current position using a FIFO queue. The BFS is presented in Algorithm 5.

Algorithm 5: BFS-based Multi-UAVs search planner

Input: Grid Cells $C_{x,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Initialize UAV positions; set all cells as unvisited; initialize empty queue Q
While the termination criterion is False, **do**
1 **For** each UAV in N do:
2 **If** Q is not empty:
3 Pop cell c_k from Q
4 Move UAV to c_k
5 Evaluate the objective function using (4)
6 Mark c_k as visited
7 **End**
8 **End**
9 **End**

2. Monte carlo tree search

Monte Carlo Tree Search (MCTS) is a planning algorithm that interweaves the exploration and exploitation processes through an independent search tree, performed via random simulations (rollouts). Particularly, this algorithm has received attention in robotics [60,61],

Table 4
Simulation parameters.

No.	Parameter name	Value
1	Mission area (m^2)	450×450
2	Cell size ($m^2 \times m^2$)	90×50
3	Grid size (cell × cell)	5×9
4	high-altitude level (m)	25
5	low-altitude levels (m)	10
6	UAV Speed (m/s)	5
7	Camera FOV (rad)	2.0
8	Camera resolution (H × W)	640×360
9	Camera direction (deg)	90° down-face
10	PSO parameters	$w = 0.9, c1 = c2 = 1.4$
11	CS parameters	$p_\alpha = 0.5$, $\beta = 2$, $a_s = 0.3$
12	ACO parameters	$\beta = \alpha = 0.5, \rho = 0.95$
13	SFO parameters	$w = 0.5, \psi_1 = \psi_2 = 1.5$
13	Human thermal threshold	>= 0.6
14	YOLO confidence threshold	>= 0.25
15	YOLO minimum box area (pixels)	64

game theory, and planning under uncertainty. MCTS is suitable for multi-step look-ahead schemes where it is necessary to consider future states/outcomes when making decisions. Specifically, individual UAVs employ MCTS to consider alternative action paths over a look-ahead horizon by simulating future actions and logging the associated payoff (e.g., the expected victim detection measure). The MCTS process is outlined in Algorithm 6.

Algorithm 6: MCTS-based Multi-UAVs search planner

Input: Grid Cells $C_{x,j}$, UAV swarm size N
Output: Next-cell waypoint for each UAV.
Initialization: Initialize UAV positions; set all cells as unvisited; initialize an empty queue Q
While the termination criterion is False, **do**
1 **For** each UAV in N do:
2 Select a promising node using UCB.
3 Generate child nodes for possible moves.
4 Simulate random rollout to estimate reward.
5 Update node values with simulated reward.
6 Move the UAV to the best child node.
7 **End**
8 **End**

Table 5
Summary of the four scenarios considered in the study.

Scenario	Altitude	Drones	Targets				Detection
			Label	Type	Location	Temp.	
1	Single	3	T1	Hum.	[37.4133708, −121.9978562]	38 °C	Thermal
			T2	Hum.	[37.4138050, −121.9956953]	37 °C	
			T3	Hum.	[37.4123595, −121.9986898]	39 °C	
2	Single	3	T1	Hum.	[37.4133708, −121.9978562]	38 °C	Thermal
			T2	Hum.	[37.4138050, −121.9956953]	37 °C	
			T3	Hum.	[37.4123595, −121.9986898]	39 °C	
			T4	Anim.	[37.4129450, −121.9956752]	36 °C	
3	Dual	4	T1	Hum.	[37.4133708, −121.9978562]	38 °C	Thermal
			T2	Hum.	[37.4138050, −121.9956953]	37 °C	
			T3	Hum.	[37.4123595, −121.9986898]	39 °C	
			T4	Anim.	[37.4129450, −121.9956752]	36 °C	
4	Dual	4	T1	Hum.	[37.4133708, −121.9978562]	38 °C	Thermal + Yolo
			T2	Hum.	[37.4138050, −121.9956953]	37 °C	
			T3	Hum.	[37.4123595, −121.9986898]	39 °C	
			T4	Anim.	[37.4129450, −121.9956752]	36 °C	

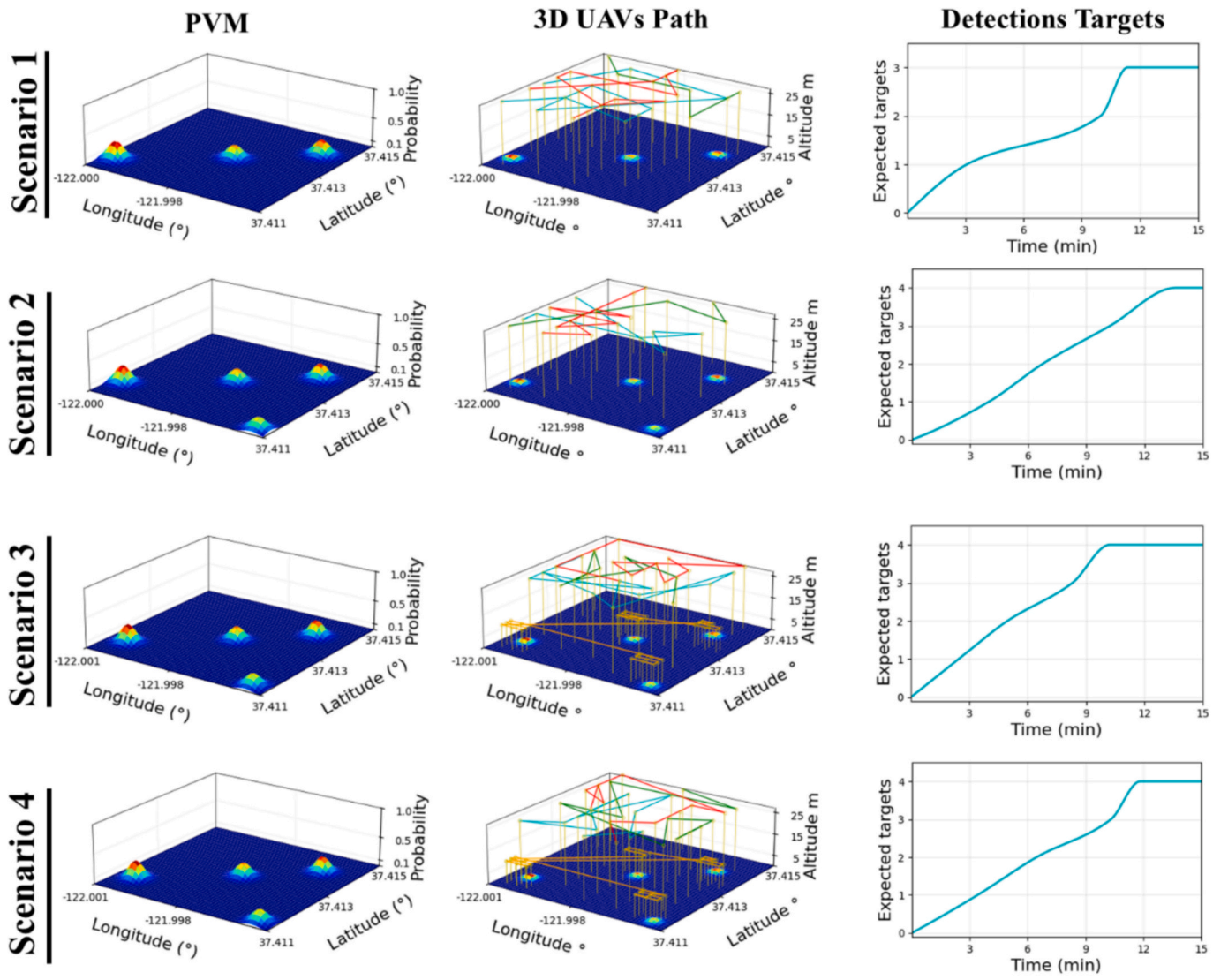

Fig. 13. Comparative visualization of the SFO-based planner for four SAR scenarios.

3.3. Victims detection and location estimation

Accurate victim detection and precise localization constitute critical requirements for the success of UAV-based SAR operations. The proposed framework, as shown in Fig. 6, addresses these requirements through a two-stage hierarchical process that integrates thermal prescreening with deep learning-based visual confirmation, followed by geodetic position estimation.

In the first stage, high-altitude UAVs equipped with thermal cameras conduct a coarse scan of the search grid. Each grid cell is assigned a thermal confidence score, which reflects the likelihood of human presence based on pixel-level temperature distributions. Next, the victim's detection probability based on temperature is modeled using a smoothed top-hat function [62], denoted as $P(T; L, Tc, W, S)$, where T represents the measured temperature, T_C is the center temperature corresponding to typical human body temperature, W is the width of the flat detection region, and S is the smoothing width applied at the boundaries. The function is defined by (14):

$$P(T;L,\ T_c,\ W,\ S) = \begin{cases} H(\frac{T - T_{min}}{S}; L), T \in [T_{min} - S, T_{min}] \\ 1, T \in [T_{min},\ T_{max}] \\ H(\frac{T_{max} - T}{S}; L), T \in [T_{max}, T_{max} + S] \end{cases} \quad (14)$$

where the parameters T_{min} and T_{max} are defined as given in (15):

$$T_{min} = T_C - \frac{W}{2}, T_{max} = T_C + \frac{W}{2} \quad (15)$$

The function is a smooth transition function, typically based on the error function [62], and is expressed as (16):

$$\mathscr{H}(y; L) = \frac{1}{2}[1 + \mathrm{erf}(\frac{L.y}{\sqrt{1 - y^2}})] \quad (16)$$

Here, L is a shape parameter that controls the sharpness of the transition between the flat region and the tails, and erf(·) is the standard error function.

Cells with confidence scores exceeding a predefined threshold are designated as candidate regions for further analysis. While this stage provides rapid, wide-area coverage and minimizes the risk of overlooking potential victims, its effectiveness is limited by the sensor

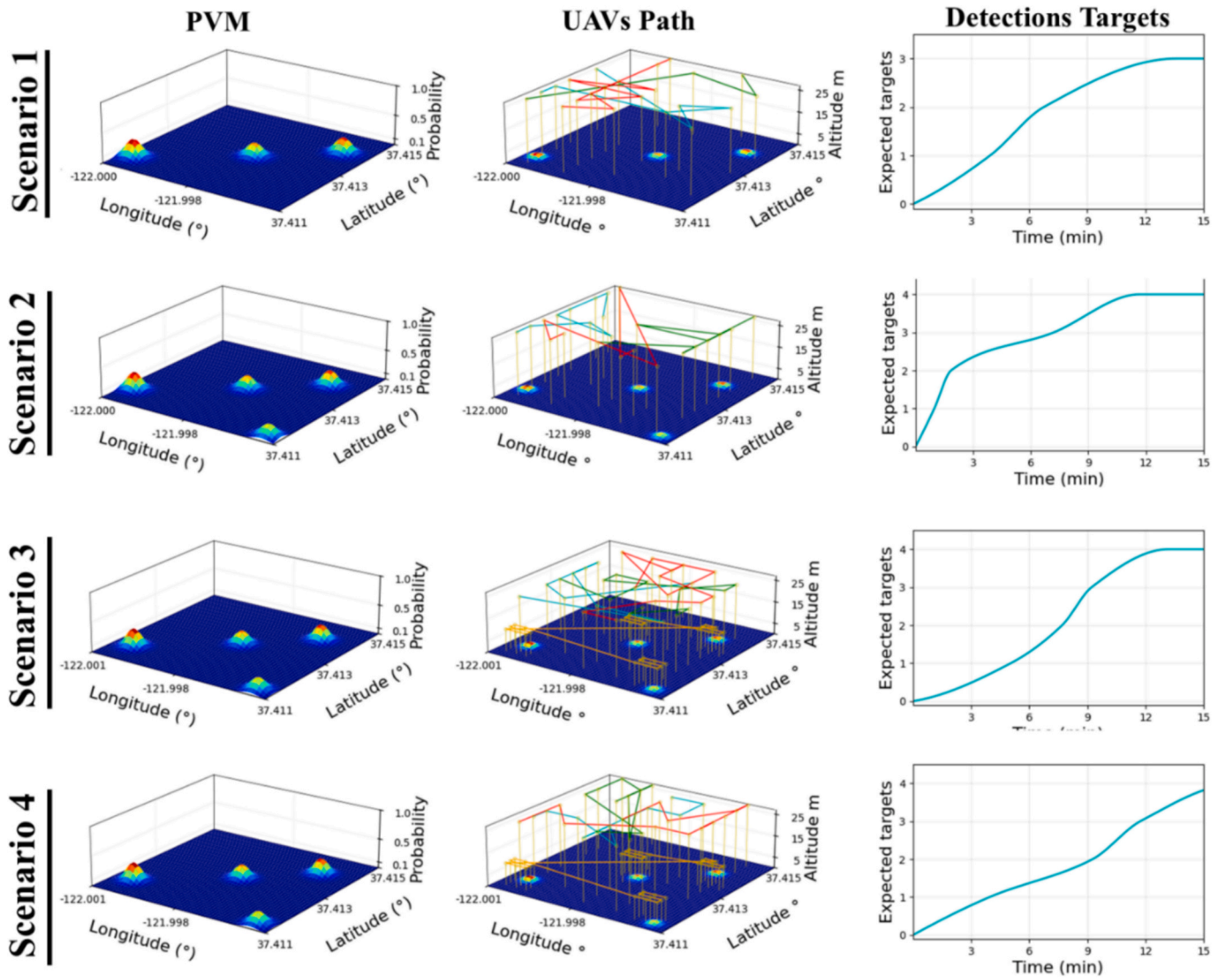

Fig. 14. Comparative visualization of the PSO-based planner for four SAR scenarios.

resolution at altitude and the susceptibility to false positives caused by non-human heat sources.

The second stage involves deploying a low-altitude UAV to refine candidate regions identified during thermal scanning. At this stage, high-resolution visual sensing is employed to refine detections and reduce the likelihood of false positives caused by non-human heat sources. More precisely, each selected cell is subdivided into smaller sub-cells according to the UAV's field of view at 10 m altitude, thereby providing enhanced spatial granularity. We employ LEAF-YOLO [63], which is a lightweight YOLO-derived detector engineered for real-time small-object performance in UAV imagery. The model was trained using the Search and Rescue Dataset (SARD) [64]. SARD is a publicly available multi-modal dataset designed specifically for aerial SAR applications. The dataset comprises real-world images captured by a high-resolution camera mounted on a DJI Phantom 4A drone, simulating realistic disaster environments. The LEAF-YOLO-N network initializes without using the weights of the pre-trained model. Model training completes when the loss function is smooth. The specific hyper-parameter information of the model used for training is shown in Table 2. Through iterative training cycles and validation testing, we achieved our target 41% mAP while implementing fairness measures to ensure equitable performance across demographic groups and environmental scenarios. The model specifications and performance are detailed in Table 3.

$$\lambda \begin{bmatrix} u_c \\ v_c \\ 1 \end{bmatrix} = K[R|t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \tag{17}$$

Once YOLO bounding box centroid confirms a victim, (u_c, v_c) is extracted and treated as the reference point for localization. This point is projected from the image plane to the ground plane using the calibrated camera model and UAV pose data, where the homography relation in (17) maps image coordinates to local ENU positions under the assumption that Z = 0, where $[X \quad Y \quad Z]$ corresponds to the ground-plane coordinates in the UAV's local ENU frame.

The resulting offsets (E, N) are then converted into geodetic coordinates relative to the UAV's current GPS position $(\varnothing_o, \lambda_o)$ using the spherical Earth approximation (18):

$$\phi = \phi_o + \frac{N}{R_\oplus}, \quad \lambda = \lambda_o + \frac{E}{R_\oplus \cos\phi_o} \tag{18}$$

where $R_\oplus$ denotes the mean Earth radius. The resulting coordinates (ϕ, λ) are logged as the estimated victim location and transmitted to the ground control station. This process transforms visual detections into actionable latitude–longitude estimates, ensuring that the detected

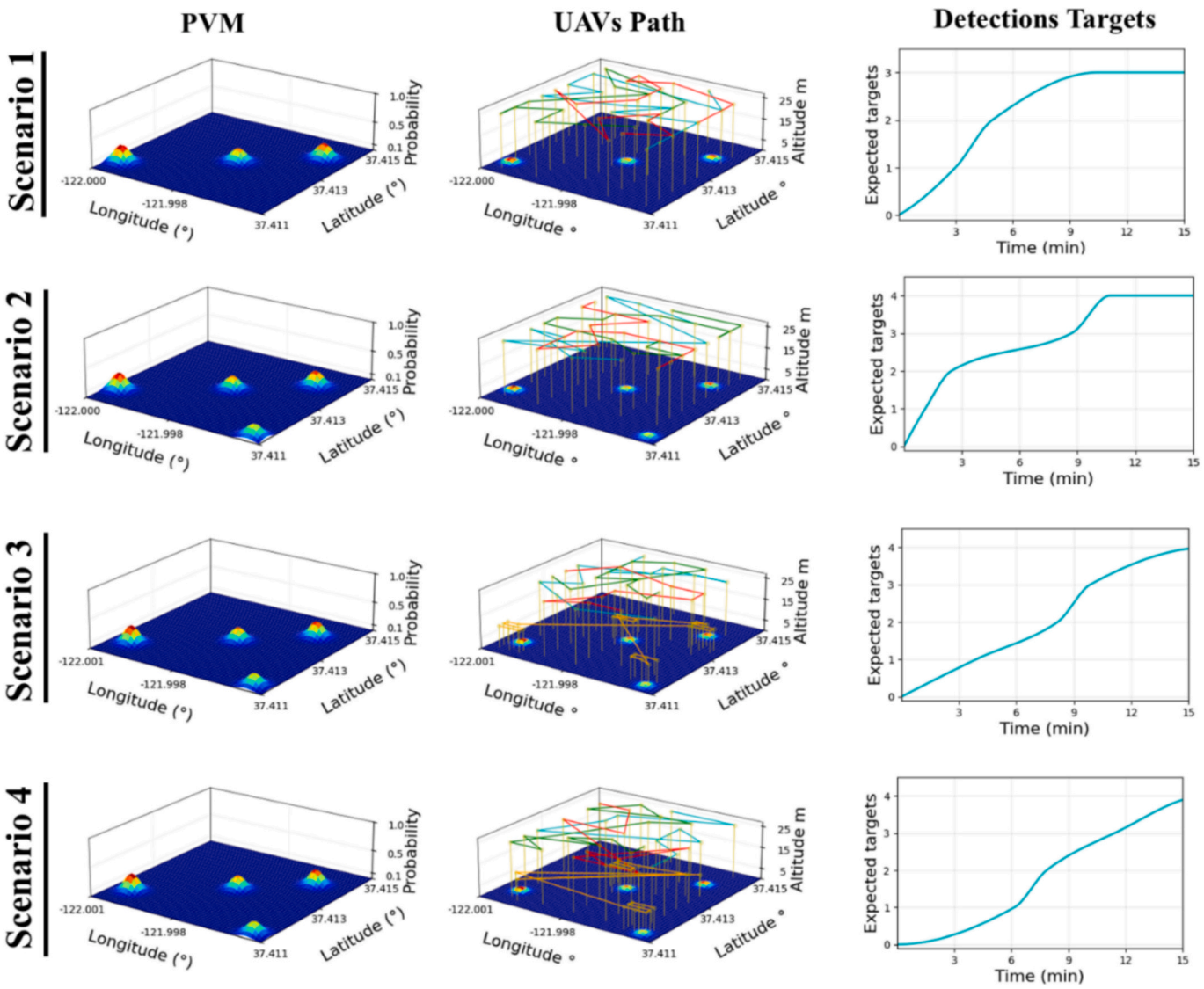

Fig. 15. Comparative visualization of the ACO-based planner for four SAR scenarios.

victims are accurately geolocated for subsequent rescue operations.

3.4. Bio-inspired search optimization with dual-altitude strategy (BSODS)

The proposed bio-inspired search optimization with dual-altitude strategy (BSODS) integrates swarm intelligence principles with a hierarchical UAV-based search framework to achieve efficient victim detection in SAR operations. Unlike conventional single-altitude or purely thermal-based approaches, BSODS leverages a coarse-to-fine search mechanism, in which high-altitude UAVs perform rapid thermal scanning to identify candidate victim-suspected cells. Furthermore, low-altitude UAVs refined these detections through vision-based confirmation using YOLO.

At the Ground Station level, the mission is initialized by deploying N UAVs across the search grid area (SGA). The ground station maintains the PVM, which is dynamically updated after each high-altitude scan. Specifically, the PVM integrates the degree of victim (DOV) probability values estimated by thermal analysis at altitude, thereby forming a probabilistic heatmap of suspected victim locations. Cell allocation is governed by a bio-inspired optimization process, modeled using the PSO update rule (4), which continuously updates the personal best (P_{best}) and the global best (G_{best}) solutions based on fitness evaluation. This step ensures that the UAVs are adaptively assigned to unexplored or high-priority regions, thereby minimizing redundant coverage while maintaining compelling exploration.

Within the High-Altitude UAV layer, each drone operates at an altitude of 25 m to scan the assigned cells. Subsequently, thermal imagery is processed to compute the DOV probability (8), which quantifies the likelihood of a human presence in the scanned region. This probability is directly incorporated into the PVM and contributes to the fitness evaluation (7), which balances detection confidence with flight efficiency. Cells with confidence values above a threshold ($\geq$0.6) are flagged in the PVM and prioritized for further investigation. The swarm-based loop of scanning, updating the PVM, and reallocating UAVs continues until either mission coverage or maximum mission time constraints are reached. In the Low-Altitude UAV layer, drones are dispatched to high-confidence cells identified in the PVM. These regions are subdivided into k sub-cells for fine-grained inspection. At an altitude of 10 m, YOLO-based object detection is applied to RGB or IR imagery for human recognition. When a victim is confirmed, the image-plane coordinates are projected into ground-plane coordinates and converted into GPS positions (12). These validated victim locations are then transmitted back to the Ground Station, where they are stored for mission reporting. If no victim is detected, the UAV proceeds with scanning the remaining sub-cells until the refinement is complete. The overall workflow of BSODS is shown in Fig. 7, which illustrates the

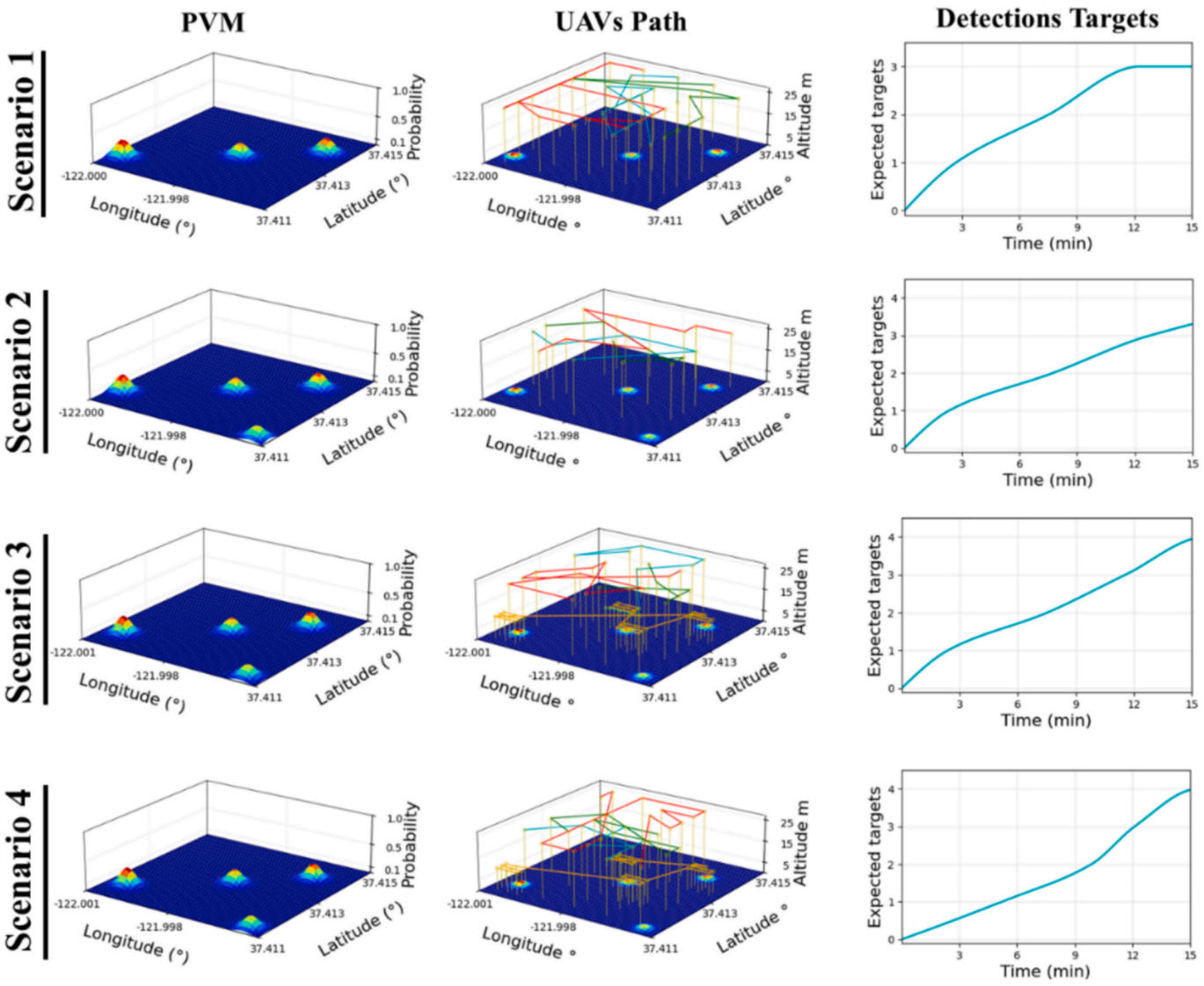

Fig. 16. Comparative visualization of the CS-based planner for four SAR scenarios.

interaction among Ground Station control, high-altitude UAV thermal mapping, and low-altitude refinement. By integrating swarm-based path planning with the continuously updated PVM, BSODS achieves a robust trade-off among search efficiency, coverage optimization, and detection accuracy. Moreover, a watchdog time is adopted to detect any communication malfunctions among the GS, HAD, and LAD.

4. System implementation

To establish proof of concept to verify feasibility, robustness and real-world applicability of the proposed dual-altitude multi-UAV SAR framework, the proposed system was instantiated and evaluated on high-fidelity simulation as well as on real hardware implementation. This dual approach to implementation allows for control over the benchmarking of an algorithm and also allows for operational readiness in real-world conditions.

4.1. Simulation implementation

Several key requirements guide the development of the simulation framework for the proposed BSODS strategy. First, the simulated disaster scenarios must closely resemble real-world SAR conditions, such as collapsed structures or obstructed urban environments, to ensure that the UAVs are tested under realistic operational challenges. Second, the platform should provide an intuitive and user-friendly mission interface that allows researchers to easily launch, monitor, and manage UAVs within the virtual environment. Third, all data collected from onboard sensors, including cameras, GPS, and telemetry, must be recorded in standardized formats to enable reliable integration, analysis and interpretation of multi-modal data for victim detection and localization. Lastly, the system must accommodate the dynamic incorporation of homemade modules, such as the bio-inspired dual-altitude optimization approach, PVM updating, and YOLO-based object detection, so that BSODS can be rigorously tested against substitute strategies under the same environment.

The proposed simulation framework integrates multiple software and hardware-in-the-loop modules to enable realistic evaluation of the BSODS dual-altitude search strategy. The overall workflow, illustrated in Fig. 8, is structured in layered form, spanning from low-level flight dynamics to high-level optimization and visualization.

At the simulation and control layer, aerial platforms and disaster environments are modeled in Gazebo, which provides a physics-based three-dimensional environment with obstacles, debris, and temperature-based victim surrogates. The UAVs are controlled by the PX4 software-in-the-loop (SITL) autopilot, enabling realistic execution of flight primitives such as take-off, hovering, waypoint navigation, and

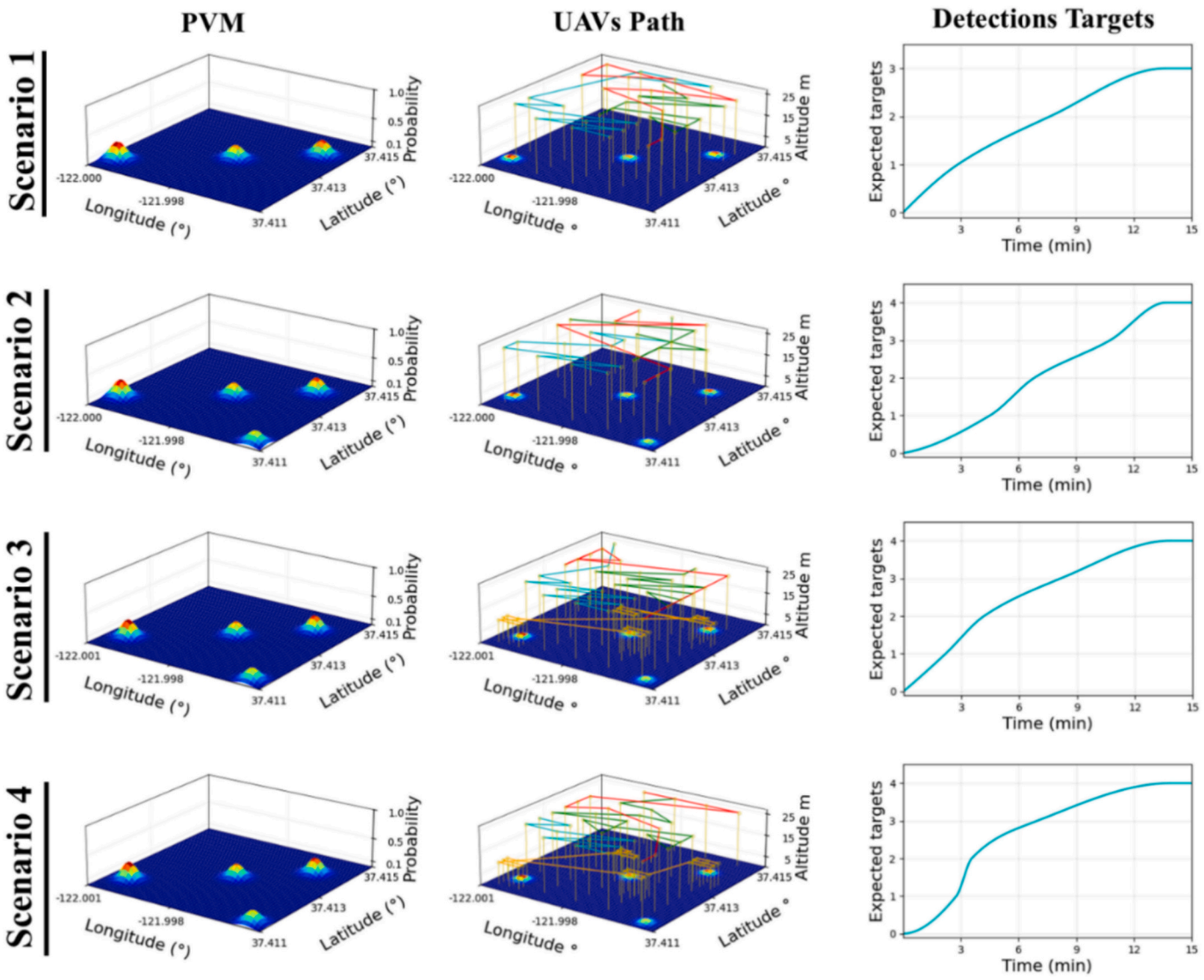

Fig. 17. Comparative visualization of the BFS-based planner for four SAR scenarios.

landing. This configuration ensures that both kinematic and dynamic behaviors are consistent with real-world operational conditions.

At the communication layer, UAVs exchange commands and telemetry with the Ground Station through MAVROS/MAVLink, while the mavsdk_server acts as the middleware bridge that enables concurrent communication with multiple UAVs. In this implementation, mavsdk_server facilitates the simultaneous control of four drones, allowing the Ground Station to dispatch mission commands, manage telemetry, and synchronize UAV operations through Python MAVSDK APIs. This step ensures that the swarm-level task allocation from the BSODS optimizer can be executed reliably in parallel across all vehicles.

At the algorithmic layer, the BSODS optimization module governs the cooperative search process. The PVM is iteratively updated using thermal observations from high-altitude UAVs. At the same time, the optimizer dynamically allocates the UAVs to maximize the victim detection probability and minimize redundant exploration. Task allocation is executed via MAVSDK in Python, which provides programmatic interfaces for mission execution and multi-UAV coordination. This modular design facilitates systematic evaluation of swarm-based decision-making under dynamic and uncertain conditions.

At the perception layer, Gazebo provides simulated multi-modal sensors, including cameras and time-of-flight cameras. In addition, video streams are transmitted using GStreamer pipelines, which provide low-latency video transport for real-time monitoring and image-based victim detection. In parallel, TOF sensor measurements are conveyed through the telemetry channel via MAVLink. These data streams are processed in real-time within the simulation, with thermal outputs encoded into victim-likelihood values, and RGB images passed to the detection pipeline. Sensor outputs are logged in standardized formats and relayed through MAVROS to the Ground Station, where they are integrated into the optimization and visualization modules. This setup replicates the dual-altitude sensing and victim-confirmation pipeline described in the methodology, while ensuring reproducible testing and evaluation under controlled simulation conditions. At the application layer, operator interaction is supported through QGroundControl and OpenCV-based visualization modules. QGroundControl provides mission configuration, telemetry monitoring, and trajectory supervision. At the same time, OpenCV is employed to display the real-time video retrieved from the UAVs, including annotated outputs from the victim-detection pipeline. This step provides both raw sensor data feed and detection report access to the human operator, hence improving transparency and situational awareness throughout the mission.

The data flow throughout the framework is a closed-loop architecture: (i) top-down control spreads from mission planning and BSODS optimization by MAVSDK and MAVROS to PX4 and the UAV airframe within Gazebo; (ii) bottom-up feedback from sensors by way of DOV

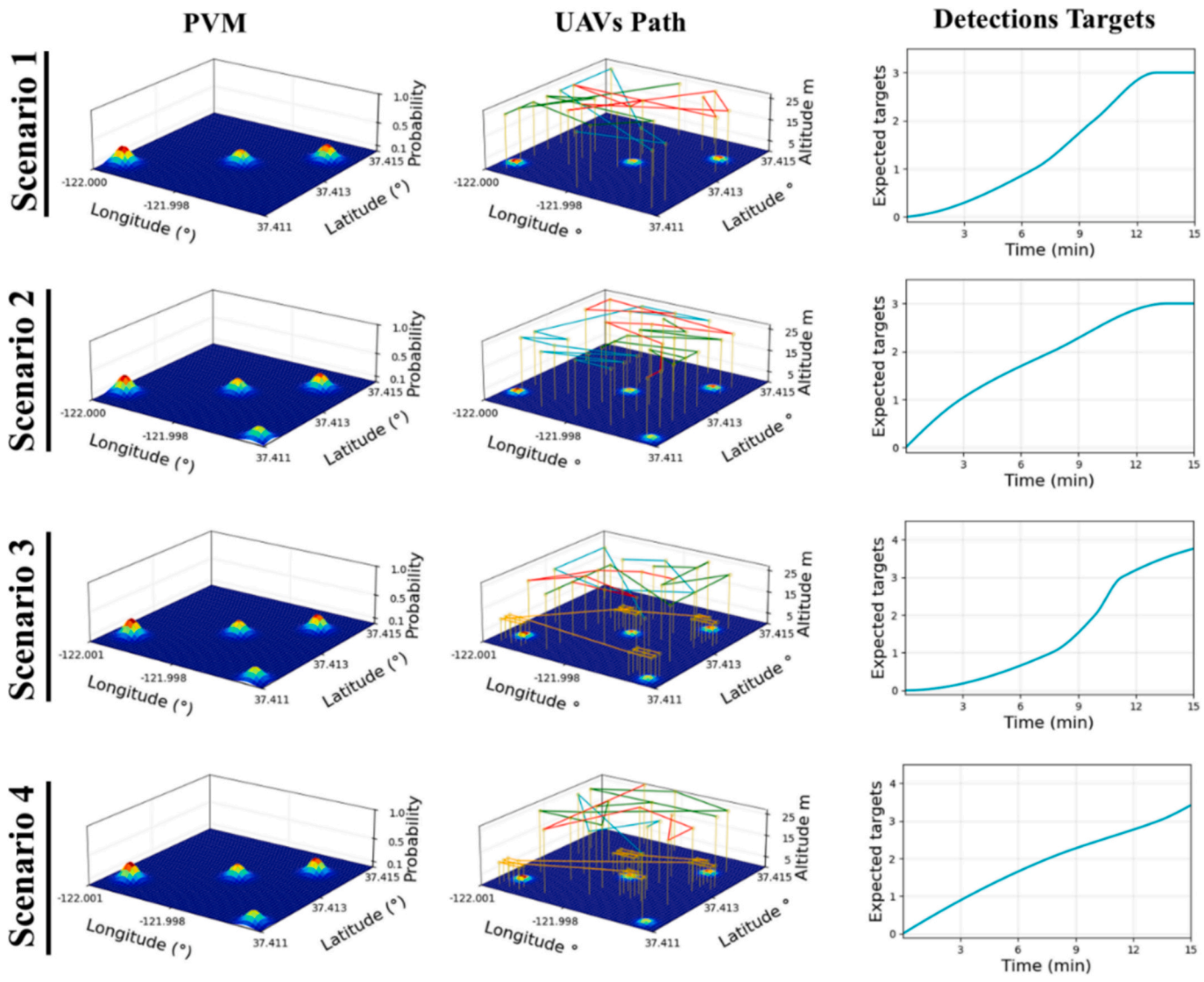

Fig. 18. Comparative visualization of the MCTS-based planner for four SAR scenarios.

estimation and YOLO detection feeds into the PVM, which informs subsequent PVM optimization cycles; and (iii) dual-altitude processes done in parallel ensure global thermal coverage by high-altitude UAVs, while low-altitude UAVs detail detections to ensure precise victim localization. The iterative loop repeats until the mission term conditions are met, including coverage completion or the expiration of mission time boundaries.

Through this modular and hierarchical architecture, the proposed simulation framework achieves equilibrium among physical reality, algorithmic versatility, and operational clarity. It supports systematic performance testing of the BSODS strategy under varied environmental conditions, providing a practical testbed for the development of UAV-based SAR systems.

4.2. Hardware implementation

This section introduces the hardware components chosen for the implementation of the proposed UAV-based SAR system. While considerable validation of the system has been achieved through simulation, a detailed description of the hardware architecture is necessary to support the argument for the practical applicability and feasibility of the proposed methodology in a real-world environment.

Fig. 9 shows the hardware architecture of the UAV-based SAR system, which consists of two types of unmanned aerial vehicles operating in different altitude regimes, i.e., high-altitude drones and low-altitude drones, and a centralized ground station. The design is optimized to support the search activities with autonomous working capabilities, real-time data transmission, and operator supervision. In the proposed framework, the flight control and mission control of high-altitude UAVs are decoupled to improve system robustness and modularity. A Pixhawk flight controller is used for controlling flights at low levels, such as stabilization, navigation, and autonomous execution of missions based on GPS data. A Raspberry Pi Companion computer is used for mission-specific thermal sensing, directly interfaced with the thermal camera for the acquisition and pre-processing of data. Communication between the Raspberry Pi and the Pixhawk is established through a UART connection, which connects the UART pins on the Raspberry Pi to the TELEM2 port on the Pixhawk. This arrangement allows for the transfer of mission commands and status information in synch but has a clear separation of the flight control and sensing tasks. An example of a high-altitude UAV search system is shown in Fig. 10. The low-altitude UAV platform features an architecture like that of a selectable UAV flight control system. Still, it is accompanied by a more powerful companion computer for intensive visual processing tasks. As is the case with the high endurance configuration, a Pixhawk flight controller manages the core flight functions. Mission control, however,

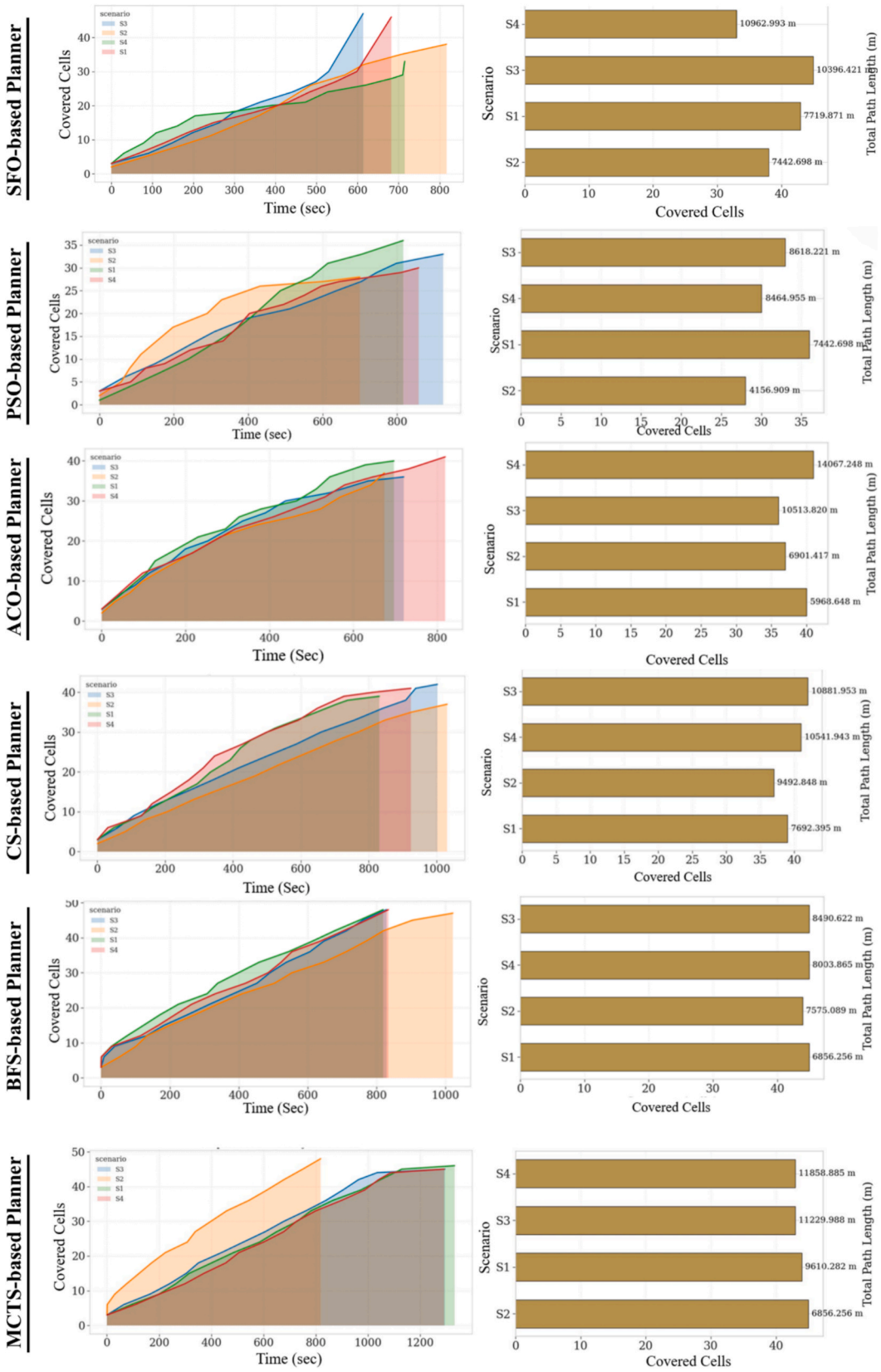

Fig. 19. Coverage and path efficiency across scenarios: (left) covered cells over mission time; (right) total UAV path length for complete coverage.

Table 6
Mission time and path length comparison of the bio-inspired and traditional planners.

Scenario	Traditional planner				Bio-inspired planner							
	MCTS		BFS		CS		ACO		PSO		SFO	
	Time (min)	Plath length (Km)	Time (min)	Path length (Km)	Time (min)	Path length (Km)	Time (min)	Path length (Km)	Time (min)	Path length (Km)	Time (min)	Path length (Km)
1	12.6	9.61	12.6	6.85	11.9	7.69	10.6	**5.96**	10.2	7.44	**10.1**	5.7
2	12.2	6.85	13.6	7.57	18.6	9.49	11.5	6.91	**11.4**	**4.15**	12.3	6.2
3	16.8	11.229	13.7	8.51	16.2	10.54	15	10.51	13.3	8.31	**10.2**	**7.92**
4	18.2	11.85	12.1	8.49	18.13	10.88	15.4	14.06	14.8	8.46	**11.7**	**8.1**

Table 7
Expected victims' results for the four scenarios.

Scenario	Expected victims				
	Count.	Label	ET	DS	Estimated Location
1	3	T2	Hum.	True	[37.413494403, −121.997476370]
		T3	Hum.	True	[37.412928141, −121.996867440]
		T1	Hum.	True	[37.414839422, −121.999085237]
2	4	T2	Hum.	True	[37.413494403, −121.997476370]
		T3	Hum	True	[37.412928141, −121.996867440]
		T1	Hum	True	[37.414839422, −121.999085237]
		T4	Hum	False	[37.412880100, −121.995914500]
3	4	T2	Hum.	True	[37.413494403, −121.997476370]
		T3	Hum	True	[37.412928141, −121.996867440]
		T1	Hum	True	[37.414839422, −121.999085237]
		T4	Hum	False	[37.412880100, −121.995914500]
4	4	T2	Hum.	True	[37.413494403, −121.997476370]
		T3	Hum	True	[37.412928141, −121.996867440]
		T1	Hum	True	[37.414839422, −121.999085237]
		T4	Anim.	True	[37.412880100, −121.995914500]

is delegated to a Nvidia Jetson Nano companion computer, which features an onboard GPU for performing real-time visual inference. The Jetson Nano processes visual information from an RGB camera connected via the CSI interface, enabling the rapid implementation of the YOLO object-detection algorithm. Communication between the Jetson Nano and the Pixhawk is also established using a UART connection, ensuring synchronization between the exchange of control signals and detection results. An example of a high-altitude UAV search system is presented in Fig. 11.

Communication between all the UAVs and the ground station is done through telemetry radios operating on 915 MHz. At the ground station end, several telemetry receivers are connected by a USB hub to a control computer (laptop). This setup would enable simultaneous communication with multiple UAVs, allowing for real-time monitoring, data logging, and mission supervision. The ground-station-interface offers visualization of UAVs' positions, mission status, and sensor data received, thereby enabling good human-robot interaction.

5. Experimentation and evaluation

To test the performance of the BSODS framework, experiments were conducted within a Gazebo-based SAR world created to simulate a real-world geographic environment. The environment is emulated from Sunnyvale Baylands Park, California, USA, which was selected for its open fields, trails, and clusters of trees, representing realistic disaster search type settings.

5.1. World deployment

The simulated SAR world, shown in Fig. 12a, reproduces key environmental features of Sunnyvale Baylands Park, including pathways, parking areas, vegetated regions, and open grassy fields. These elements were carefully aligned with the real-world setting to ensure geographic fidelity. Moreover, Fig. 12b shows the corresponding 3D perspective view from Google Earth, while Fig. 12c provides the high-resolution satellite imagery of the park from Google Maps. A schematic map representation is shown in Fig. 12d, highlighting trails and landmarks that were also incorporated into the Gazebo model. By integrating these multiple sources, the simulation environment achieves both realism and consistency with real-world coordinates, enabling the controlled placement of victim surrogates. Victims were positioned at both exposed and obscured locations, some in open areas to represent easily detectable cases, and others near trees or structural features to test the robustness of the dual-altitude detection strategy.

5.2. Scenarios

The scenario settings are as follows: The Byland world was featured as a setting, which is a realistic outdoor environment suitable for SAR operations. The mission site was 450 × 450 m with a grid size of 5 × 9 cells and a cell size of 90 × 50 m; two UAVs worked simultaneously: one at an altitude of about 25 m for thermal imaging, and the other flew at about ∼10 m altitude to fly under foliage for low-altitude visual confirmation. Every UAV was equipped with a simulated thermal camera (FOV: 2.0 rad, resolution: 640 × 360) and an RGB camera processed by YOLOv5, whose detection confidence thresholds were higher than or equal to 0.25. In particular, the heat detection threshold value was determined to be greater than 0.6. The coordination of the UAVs was achieved using the PSO algorithm, with performance indicators applied to optimize a fitness function that also served as a trade-off between detection probability and travel distance.

All simulations were run on a high-end PC with an Intel Core i9-12900 K CPU (16 Cores, 24 threads, 3.2 – 5.2 GHz boost), 64 GB RAM DDR5, and an NVIDIA RTX 3090 GPU (24 GB GVRAM) for Ubuntu 22.04 LTS. These nominal conditions ensure that all experiments are conducted under a well-defined and comparable structure, thereby allowing for a fair comparison of distinct mission configurations. In the context of this standard setup, four different setups are specified to evaluate the performance in relation to how changing altitudes, types of sensing, and distributions of victims affect performance. The parameters used in the simulations are listed in Table 4.

- **Scenario 1:** In this case, a single-altitude search strategy is employed, where three UAVs operate at the same altitude while relying exclusively on thermal sensing. The search area contains three victims, all represented as persons with human thermal

signatures. This scenario provides a baseline for evaluating the performance of multi-UAV thermal search in a simplified setting with homogeneous targets.

- **Scenario 2:** This scenario also applies a single-altitude thermal search strategy with three UAVs, but the number and type of victims differ. Four victims are placed in an environment consisting of three persons and one animal that exhibits the same thermal characteristics as those of a human. This setting introduces an additional challenge by incorporating a non-human target that can generate potential false detections.
- **Scenario 3:** In this setup, a dual-altitude search strategy is adopted. Three UAVs operate at high altitude to conduct wide-area coverage, while an additional UAV flies at a lower altitude to perform refinement around detected hotspots. Thermal sensing remains the sole detection method, and the victim set consists of three persons and one animal. This configuration reflects a more advanced cooperative search strategy where altitude differentiation enhances both coverage and accuracy.
- **Scenario 4:** The final scenario also applies a dual-altitude search strategy with three UAVs at high altitude and one UAV at low altitude. Unlike Scenario 3, however, detection is performed using a combination of thermal sensing and YOLO-based computer vision, thereby enabling multimodal target recognition. The victim set again includes three persons and one animal. This scenario represents the most complex setup, combining cooperative multi-altitude coordination with hybrid sensing for robust victim identification.

The four experimental scenarios were designed to gradually increase operational complexity, starting with a baseline single-altitude thermal search involving only human victims. Successive scenarios introduced mixed victim types, dual-altitude strategies, and multimodal detection methods. The defining parameters of each scenario are systematically summarized in Table 5.

6. Results and discussion

This section discusses the performance analysis of the designed UAV search framework under various mission scenarios. Particularly, the multi-UAV SAR system has been tested in four progressively complex scenarios, under six planning techniques: SFO, PSO, ACO, CS, BFS, and MCTS. The comparison was based on mission time, trajectory efficiency, detection performance, and semantic congruence. These findings are illustrated in Figs. 13–19 and quantitatively compared in Tables 6-7, which present both the system-level overall performance and the individual victims' classifications resulting from the fusion process.

Figs. 13–18 present a detailed visual analysis comparing the six planners (SFO, PSO, ACO, CS, BFS, and MCTS) across the four SAR scenarios. The behavior of each planner is depicted in three complementary visualizations for every scenario: (i) the PVM, which visualizes the map over space for potential targets; (ii) UAV trajectory plots representing the spatiotemporal evolution of drone paths; and (iii) detection curves capturing success and temporal ordering of identifying victims. These scenario-specific qualitative evaluations are also underpinned by Table 6, which shows the average mission time and total flight distance values for each planner. In addition, Fig. 19 shows a cross-scenario analysis, combined over all the scenarios: (i) on the left, we report area coverage as a function of time (cumulative number of grid cells processed), and (ii) on the right, we compare the total path length required by the UAVs to achieve an end-to-end full coverage. Taken in sum, these results provide localized and system-wide insights into planner efficiency, coordination robustness, and detection reliability.

Scenario 1, with three separate human targets in a low-complexity environment, was considered the baseline. All planners achieved a 100% detection rate with no false positives. In contrast, the SFO generated the best mission performance in terms of time efficiency (averaging 10.1 min from and to the base). As presented in Table 6, the predicted victim positions obtained by the SFO showed low spatial error for thermal images taken from high altitudes only. The PSO, ACO and CS fan-out achieved similar detection results, albeit with slightly longer flown paths, due to relatively looser inter-UAV cooperations. The BFS revisits cells due to its BFS-based planning, and MCTS accounts for latencies through its sampling navigation.

Scenario 2 increased ambiguity by introducing a thermally deceptive animal target (T4) alongside humans. As shown in Table 7, all the planners correctly labeled T1–T3; however, the PSO returned the minimum flight paths and mission time, yet carried out complete detection. However, all non-semantic planners (SFO, PSO, ACO, CS, BFS, and MCTS) considered T4 as human, highlighting the fundamental weakness of thermal-only modalities. Despite this outcome, the PSO performed better than the other algorithms by adaptively focusing on high-probability regions and reducing false trajectory overlap, as shown in Fig. 11. The BFS is exhaustive but not efficient in terms of spatial distribution. All SFO, ACO and CS models only provided medium performance, and there was no way to correct the paths in real-time to prevent them from becoming stuck. The MCTS exhibited unstable convergence, indicating that semantic enhancement is necessary under complex thermal profiles.

Scenario 3: We introduced double-altitude tuning for low-altitude UAVs to confirm high-probability areas identified by the higher-altitude scouts. This configuration had a positive impact on localization, and the SFO emerged as the best option again. As indicated in Table 7, the victim coordinates estimation was better with the SFO compared to T4, which remained misclassified due to the thermal sensing alone. An adaptive inter-UAV role assignment algorithm for coordinated search and verification ensures that the search effort is balanced with exploitation, thereby optimizing the overall search process. The PSO, ACO and the CS considered the edge in Scenario 2 only marginally more than above; however, with no refined coordination. The BFS's pattern-based search made repeated visits, and the MCTS's stochastic rollouts were not effective at capturing semantic target clusters. Fig. 13 illustrates the difference in efficiency of the path between UAVs for the SFO, while maintaining structural homogeneity among agents. The findings verify the robustness of the SFO in multi-altitude task assignment.

Scenario 4, thermal and semantic (YOLO-based) fusion, is the most complex scheme. This time, the SFO is the only one that achieved perfect detection and classification among all targets, as well as T4, as shown in Table 6. Then, the SFO integrated thermal cues with semantic confirmation in a smooth manner, leading to the correct rejection of T4′s false positive despite its human-like biomass footprint. Although the mission duration increased (11.7 min), the SFO preserved near-optimal coverage and trajectory economy. The PSO, ACO, and the CS, all without semantic integration, misclassified T4 again. On the other hand, the BFS exhibited predictable exploration and did not show a preference for semantically hotspots. The MCTS, despite being a probability-based approach, did not utilize any semantic feedback during the decision-making process. Therefore, the integration of multi-modal sensing and swarm intelligence makes the SFO the sole method that can balance high detection with semantic reliability. The SFO planner allocates the UAV paths in a well-distributed and highly structured manner, i.e., with minimal overlap, which leads to balancing the area partitions and efficient grid exploration. Its adaptive change between exploring and exploiting phases helps reduce premature clustering and speed up convergence towards suspected victim regions. On the other hand, PSO and ACO algorithms exhibit premature convergence and repetitive retracing of the path, resulting in increased redundancy and slower coverage expansion. MCTS and BFS generate unstable and oscillatory trajectories because of frequent leader or spiral re-rotation, which increases the path length and reduces search efficiency. CS maintains moderate dispersion, but lacks directed convergence, resulting in a prolonged mission time. In summary, the SFO planner exhibits better spatial coordination, redundancy avoidance, and convergence

properties than the other planners, thus representing the most effective strategy for the dual altitude search and rescue missions among the algorithms considered. The results validate that the planner based on the SFO provides the best search efficiency overall, the fastest exit and composite exploration performance, and is the most appropriate coordination strategy for time-critical multi-UAV SAR missions.

Figs. 10–14, along with Tables 4 and 5, collectively demonstrate that the proposed PSO-based multi-UAV strategy achieves rapid coverage, efficient path allocation, and high detection reliability. Beyond these overall findings, several key advantages can be highlighted to demonstrate the strengths of the proposed framework:

- Enabling rapid coverage of large areas with efficient PSO-based trajectory allocation.
- Maintaining high detection reliability across different mission complexities.
- Dual-altitude refinement improves localization accuracy compared to single-altitude operation.
- Hybrid sensing (thermal + YOLO) enhances robustness by correctly distinguishing human and animal targets.

However, alongside these advantages, certain limitations were also observed, which outline directions for further improvement:

- The current evaluations are limited to simulation, and real-world hardware deployment is required to validate practical feasibility.
- Hybrid sensing and low-altitude refinement increase overall mission duration.
- Single-modality thermal detection remains prone to false positives in thermally cluttered environments.

7. Conclusion

In this paper, a multi-UAV system based on SFO was proposed, aiming to improve victim detection and localization in SAR through the coordination of high-altitude exploration and low-altitude refinement. The strategy successfully integrates thermal sensing and YOLO-based visual detection to alleviate the classification ambiguity issue and enhance localization accuracy. The findings demonstrated that combining sensing and altitude coordination enabled efficient coverage with reliable target detection and discrimination between human and animal objects across various mission complexities. In addition to validating the framework in a controlled simulation setting, these results highlighted the potential applicability of this approach in realistic disaster response conditions, where prompt detection and precise localization of casualties are essential. In particular, the presented architecture serves as a proof of concept for scalable multi-agent SAR systems. As future work, the current system can be extended into real-world scenarios through hardware-in-the-loop testing to evaluate the reliability of swarm communication (e.g., latency and packet loss), onboard inference performance on embedded platforms (e.g., the NVIDIA Jetson computing platform), and the synchronization of thermal and RGB streams. Furthermore, incorporating strong fail-safe strategies to address UAV drop-out, multi-agent replanning under GPS/IMU uncertainty, and adaptivity to dynamic environments (e.g., smoke, rain, or rubble-filled areas) remains an open problem.

CRediT authorship contribution statement

Abbas Aqeel Kareem: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ahmed Jabbar Abid:** Data curation, Conceptualization. **Dalal Abdulmohsin Hammood:** Data curation, Conceptualization. **Roobaea Alroobaea:** Supervision, Resources, Conceptualization. **Amr Yousef:** Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to acknowledge the Deanship of Graduate Studies and Scientific Research, Taif University for funding this work.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

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